

Title
Implementing a Multifaceted, Comprehensive, Appropriate Care Package for People with Advanced and Terminal illnesses needing ICU support and knowing its benefits on Quality of Dying, Family Satisfaction, Care Cost and Resource Utilisation.

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Glossary of terms

1. Multifaceted Comprehensive Appropriate Care Package (MCACP)

It is an evidence-based clinical care package having several features and comprehensively delivered by an interdisciplinary team of ICU and palliative care providers, which is found to be appropriate (from empiric literature) in people with advanced and terminal illnesses having a medically futile situation where continuation or escalation of ICU measures is unlikely to provide any clinical benefit. **Figure 1** and **Table 1** (below). Please **see Supplement I** for an Emergency Toolbox for Symptom Management of ICU Patients with Advanced Illness. Broad categories are A. Clinical: symptom management, family meetings, deprescribing B. Technical: data collection tools, e-learning, decision aids C. System-level: SOPs for futility, checklists, audit tools. We recognize that ICUs can operate at varying service levels. Therefore, we will include a table that outlines the core and optional components of the intervention for each ICU level. Additionally, we will define the Minimum Viable Intervention Set (MVIS). This will include core components such as a symptom kit, documentation templates, family meeting protocols, and optional components like syringe drivers, electronic health records (EHR), and simulation training.

2. People with Advanced and Terminal Illness (ICMR definition¹)

Advanced and Terminal illness refers to the clinical condition from which recovery cannot be expected with the available treatment, and death is considered to be unavoidable in the foreseeable future². It also refers to an incurable and irreversible condition caused by injury, disease, or illness that would cause death within a reasonable period of time in accordance with accepted medical standards and where the application of life-sustaining treatment would serve only to prolong the process of dying³.

3. Quality of Dying

It refers to the family's perception of the death of their loved ones in terms of overall care, communication and support, trust, respect and dignity, and symptom management, often measured using the 25-item *ICU-QODD* questionnaire⁴ (**Supplement II**). Moreover, it encompasses if the end-of-life care aligns with documented preferences (e.g., advance directives, family-reported choices) and the patient has received symptom management interventions tailored to patient comfort.

4. Family Satisfaction

It refers to family experience regarding the care received in the ICU, family concerns and needs addressed by the ICU providers, care coordination, emotional support, the ICU environment, and the overall skills of the ICU team. It is often measured using Family Satisfaction with ICU (*FS-ICU*)⁵ (**Supplement III**). It also involves qualitative exploration of family views on communication and involvement in decision-making processes and their experiences with care coordination and emotional support. We will interview the family 3–4 weeks after death, not immediately, and will utilize trauma-informed interviewers trained to handle grief sensitivity. We will triangulate FS-ICU with open-ended interviews and clinical notes.

5. Care Cost

It refers to the cost of ICU care (direct and indirect cost) in patients with Advanced/Terminal illness treated in an ICU and the potential cost-saving benefit of the intervention package (MCACP) when introduced along with standard ICU care or after discontinuation of ICU care.

6. Resource Utilization

It refers to the average length of stay in the ICU for a patient and the number of invasive procedures or non-beneficial treatments avoided due to the intervention package (MCACP).

7. End-of-Life Phase in the ICU

END-of-Life Scoring-System (*ENDING-S*) score will be used to identify the End-of-Life Phase in the ICU. *ENDING-S* Score of ≥ 12 in ICU satisfying the inclusion criteria above will be considered for EOLC.

$$ENDING-S = (7.25 \times \text{Days of MV/ICU LoS}) + (10.45 \times \text{Days of vasoactive drugs/ICU LoS}) + (3 \times \text{Sepsis}) + (0.3 \times \text{ICU LoS})$$

Where Days of MV/ICU LoS expresses the ratio between the current days in which the patient requires mechanical ventilation (MV) and the current ICU length of stay (LoS, quantitative variable), Days of Vasoactive drugs/ICU LoS expresses the ratio between the current days in which the patient requires vasoactive drugs and the current ICU length of stay (quantitative variable), Sepsis expresses a septic condition (dichotomous variable: 1 if currently affected, otherwise 0), ICU LoS expresses the current ICU length of stay (quantitative variable).

Summary**Rationale**

Intensive Care Units (ICUs) in India frequently provide invasive, unhelpful treatments to patients facing advanced and terminal illnesses. It can significantly impact the patients' quality of life in their final days. Despite the guidelines on end-of-life care established by the Indian Association of Palliative Care (IAPC) and the Indian Society of Critical Care Medicine (ISCCM) since 2005, there has been a lack of integration of palliative care with ICU services. This has resulted in unnecessary resource usage and increased financial burdens for families. There is considerable variability among Indian ICUs regarding infrastructure, staffing, training, legal status, and socio-cultural norms. Additionally, challenges to implementing any interventions are compounded by clinician behavior, institutional readiness, legal uncertainties, family dynamics, and commercial pressures. This study aims to implement and evaluate a Multifaceted, Comprehensive, Appropriate Care Package (MCACP) designed to provide ethical, patient-centred, and resource-aware care for terminally ill ICU patients. The implementation science framework acknowledges that MCACP, as described here, may not be a definitive solution but a provisional and evolving model.

Novelty

The MCACP is an evidence-based framework developed through an umbrella review (overview of 40 systematic reviews). It is the first of its kind to be operationalized and tested in Indian ICUs utilizing an implementation science framework, integrating palliative care with ICU protocols. The PI has previously developed national guidelines, including the BLUE MAPLE program, influencing national policies on end-of-life care. This study expands on prior work to assess feasibility, implementation, and impact on patients, families, and healthcare systems.

Objectives

- Co-develop the MCACP intervention through iterative, participatory design.
- Pilot the MCACP intervention in two ICUs with existing palliative care integration to refine implementation and intervention and explore the key facilitators and barriers.
- Expand implementation to four ICUs with existing palliative care access, with an aim of refining implementation and intervention and evaluating its effectiveness on quality of dying, family satisfaction, cost, and resource utilization.
- Extend implementation to four ICUs without existing palliative care access, with an aim of refining implementation and intervention, and evaluating its effectiveness on the quality of dying, family satisfaction, cost, and resource utilization.
- Develop a national technical resource hub to facilitate widespread adoption of the MCACP in India.

Methods (detailed linkage of methods with objectives is provided for each objective below)

- A mixed-methods approach that includes Delphi consensus, qualitative interviews, and observational studies.
- The use of implementation science frameworks, such as the Consolidated Framework for Implementation Research (CFIR) and Normalization Process Theory (NPT), to assess adoption and

sustainability - Integration of CFIR across the design, piloting, and evaluation phases, and utilization of NPT to inform qualitative research and realist interviewing.

- A quasi-experimental pre-post study design to measure the impact of the MCACP intervention.
- An ethical framework for in-depth interviews, including pathways for emotional support.
- Ensuring the training and standardization of qualitative skills across various sites.
- A cost analysis to evaluate the cost-effectiveness and budgetary implications of scaling up the intervention.

Expected Outcomes

- Improved quality of dying, ensuring care aligns with patient preferences.
- Enhanced family satisfaction with ICU care, decision-making, and emotional support.
- Reduction in non-beneficial ICU treatments, optimizing resource utilization.
- Cost savings due to reduced ICU stays and fewer unnecessary procedures.
- A scalable, standardized model for palliative care integration in ICUs across India.

Equity and Representation

- We will try our best to prioritize the voices of women leaders, caregivers, and staff nurses.
- Include voices of the bereaved and vulnerable families.
- Design and distribute simple decision-making tools and stories so that it can work in district-level public hospitals.

Scalability Strategy

- Define a minimum model for hospitals with low resources.
- Identify core vs optional components of MCACP.
- Develop tools to assess hospital readiness to adopt MCACP.

Keywords

Intensive care unit (ICU), advanced and terminal illness, appropriate care, palliative care, interventions

Background and Rationale

A. Why this Population and Setting?

Creating a Centre for Advanced Research (CAR) aims to benefit people with advanced or terminal illnesses in Indian ICUs or those needing ICU support because this population faces a high burden of health-related suffering, receives unhelpful, inappropriate intensive medical interventions at end-of-life, compromising their quality of life and quality of dying⁶, making them an ideal group for evaluating the impact of a multifaceted, comprehensive, appropriate care package. Furthermore, the Indian healthcare setting, particularly ICUs, lacks standardized palliative care integration, highlighting the need for an intervention that balances intensive care with patient and family-centred approaches⁷. Although there are position statements⁸⁻¹⁰ developed in India targeting this population and setting jointly by the Indian Society of Critical Care Medicine and the Indian Association of Palliative Care, these are yet to be operationalized as intervention packages. Their outcomes are yet to be assessed, necessitating the development of a CAR, which can determine the feasibility of implementing a multifaceted comprehensive appropriate care intervention package in Indian ICUs, explore its benefits, and create a resource hub for pan-India delivery.

B. Why this Intervention Package?

The Multifaceted, Comprehensive, Appropriate Care Package (MCACP) is chosen because it integrates palliative care principles with standard ICU care, ensuring a more ethical, humane, patient-centred, and resource-conscious approach. The package incorporates symptom management, anticipatory care planning, family engagement, and ethical decision-making to improve the overall care experience while addressing feasibility, acceptability, and implementation barriers in real-world ICU settings. It is designed based on robust evidence from an umbrella review conducted by the principal investigators, a systematic review of 40 systematic reviews exploring international evidence on palliative care interventions in ICUs¹¹. Multifaceted Comprehensive Palliative Care Interventions Developed from the Umbrella Review are illustrated below in **Figure 1**.

C. Why is the Appropriate Care Concept (ACC) incorporated in a CAR proposal?

The Appropriate Care Concept (ACC)¹² comprises five key components. Care should be evidence-informed, provided by a clinically skilled team, and centered around the needs of patients and their families. This approach aims to optimize resource use and promote equity in healthcare. Evidence-based care is focused on achieving favourable health outcomes while effectively incorporating research findings into practice. A proficient clinical team possesses the expertise and training to integrate expert opinions into clinical decision-making, facilitating personalized treatment for individual patients. Patient and family-centred care involves delivering responsive and coordinated services fostering patient autonomy through open communication and shared decision-making. Effective resource management aims to minimize waste, avoid unnecessary interventions, ensure efficient allocation of resources, and reduce costs. Equity encompasses the fair distribution of resources across various settings to achieve consistent health outcomes and provide non-discriminatory care¹².

D. Why these Outcomes?

The selected outcomes—feasibility of implementation¹³, quality of dying¹⁴, family satisfaction¹⁵, health economics, and resource utilization¹⁶ are essential for evaluating the impact of the intervention. Feasibility pertains to the practical application of the intervention across various ICU settings. Quality of dying

assesses whether patients receive care that aligns with their values and preferences. Family satisfaction encompasses the overall experience of care and the decision-making processes involved. Health economics and resource utilization evaluate the cost-effectiveness of the intervention¹⁷ and its potential for scalability in resource-constrained healthcare environments.

Figure 1: Multifaceted Comprehensive Palliative Care Interventions Developed from the Umbrella Review¹¹

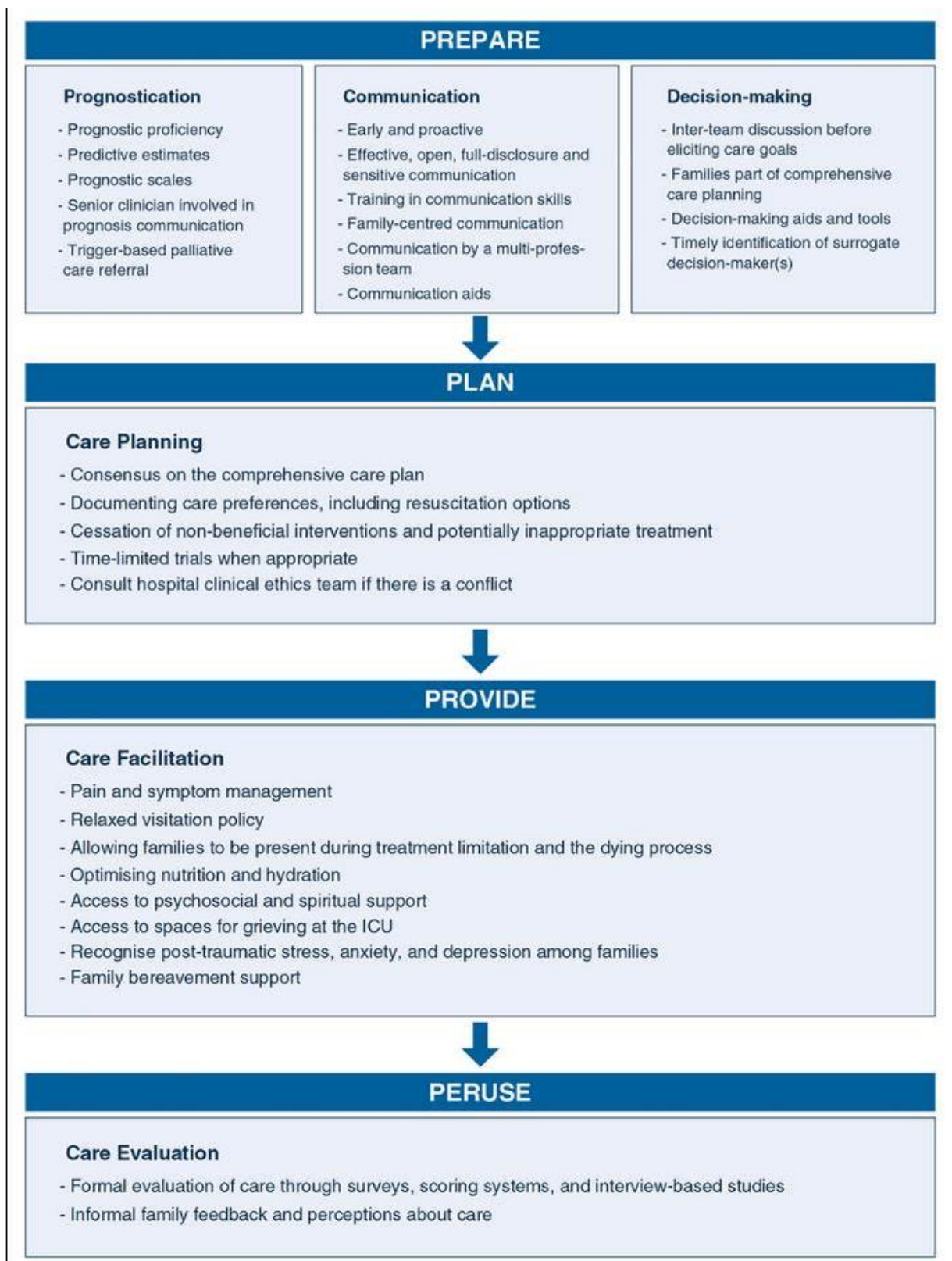


Table 1: Multifaceted, Comprehensive, Appropriate Care Package for People with Advanced and Terminal illnesses needing ICU support

Intervention	Description of the Package	Mode of Delivery
1. Prognostication	• END-of-Life ScorING-System (ENDING-S) score will be used to identify the End-of-Life Phase in the ICU. (as described before) • Co-develop user-friendly resources like ○ Clinician training manual (text and videos) for prognosis- based determination of medical futility ○ A family education manual for prognostic communication during a family meeting ○ Criteria and triggers for palliative care referral in ICUs • Pilot usability in any setting before full rollout	• Embed resources into the routine workflow with iterative testing and frontline feedback loops • Staggered workshops for ICU healthcare providers on both manuals developed for prognostic determination and communication focusing on both skill-building and ethical framing
2. Review and Consensus	• A manual for setting up the Critical Care Review Board - CCRB (team composition, meeting frequency, meeting decision-making process, implementation of CCRB outcomes), acknowledging and addressing the power dynamics in the design of CCRB • A co-designed SOP for Physician Consensus on Review of the Care Process, Endorsement of Clinical Futility, and Documentation of Futility with flexibility for tiered hospital settings • A manual for medical futility in various clinical conditions like cancer, stroke, CKD, heart failure, etc, from evidence synthesized in the literature	• Incorporation of the process of futility endorsement by the physicians in the routine ICU workflow - Adaptive workflows tested in low- and high-resource ICUs • Workshops on educating ICU providers on CCRB manual and decision-making, bedside futility endorsement process, and futility determination in advanced and critical illness including case audits with equity lens (who gets reviewed, and why) • Role-playing and simulation-based training to address real-world ambiguity
3. Communication	• Family information leaflets and brochures outlining the care process	• Incorporating family communication and documentation of family

	• Communication manuals for advanced communication scenarios in ICU settings and using existing tools to evaluate outcomes of communication (scenarios like – breaking bad news, handling conflict, handling anger, managing expectations, navigating uncertainties, communication regarding the dying process and after-death care, communicating death information, bereavement support videos) - narratives/case studies as tools for training ICU teams in empathy, uncertainty, cultural framing • Contextual Indian setting and culture-based videos on the above-mentioned scenarios • Templates for documentation of ICU communication and counselling • Translation of leaflets/manuals into regional languages • Training staff to recognize distress and provide immediate psychosocial support (Supplement IV)	communication/counselling in routine ICU practice • Workshops to ICU providers in communication skills training using the communication manuals and videos created • Peer-led workshops and scenario-based communication labs • Tools adapted for frontline staff (e.g., nurses and AYUSH practitioners in govt hospitals) – as much as possible • Feedback from bereaved families used to refine manuals and SOPs
4. Decision-Making	• Family leaflets and brochures on their rights and responsibilities in the decision-making process and FAQ document on the process involved in participating in a family meeting and the decision-making process, with tiered decision-aid tools (video + print + low-literacy graphic sheets) • Manual on conducting a family meeting that encompasses – the process of setting up the family meeting, stakeholders involved, steps in eliciting goals of care, a framework for shared decision-making, documenting family meetings and implementing outcomes of family meeting	• Incorporating multi-stakeholder family meetings as part of routine ICU care, and to include social worker if available • Workshops and training for ICU providers on conducting family meetings and all the steps involved in achieving a shared decision, in an iterative manner on shared decision-making as daily practice, not crisis-only • Workshops and training for ICU providers on using decision-making aids • Decision aids field-tested with patients/families from

	• Manual on the decision-making process and aids (text and videos) - video-based decision-support tools and verbal descriptors • Including modules that address decision-making under socio-economic stress or gender imbalance • Capture undocumented family voices via feedback sessions	different socioeconomic classes
5. Care Planning	• Manual on Comprehensive Anticipatory Care Plan (CACP) – that encompasses the process of consenting and documenting withholding/withdrawing life-sustaining measures, de-prescribing, down-scaling investigations, de-activating devices anticipatory prescribing, ceilings of therapy, place and nature of care, etc. in both digital and paper formats, with translations and illustrations for low-literacy caregivers • Manual on infrastructure, staffing, resources, policies, and external liaison support for implementing MCACP in ICUs • Co-designed standard consent templates for DNAR, de-prescribing, deactivation of devices, de-escalation of therapies • Readiness checklist for infrastructure before rolling out CACP • Design for modular use in resource-constrained ICUs	• Incorporating a Comprehensive Anticipatory Care Plan (CACP) in routine ICU care for people with advanced and terminal illnesses in the ICU • Workshops and training to ICU providers on CACP and various aspects of CACP outlined • Compliance audit of documentation of consent for CACP components, with Audit forms including field-level usability feedback • Build-in adaptation workshops every 6 months post-rollout
6. Care Delivery	• Manual for assessment and management of end-of-life symptoms (initial and ongoing) after withholding or withdrawing life-sustaining treatment in the ICUs • Formulary of essential drugs and preparations used for end-of-life symptom management and	• Incorporating symptom management protocols in routine ICU practice for people with advanced and terminal illnesses in the ICU receiving end-of-life care

	procedures involved in setting up palliative care infusions and delivery, with stepwise drug availability checklist for public sector ICUs • Framework for daily family communication for families of patients receiving end-of-life care • Framework for eliciting after-death care preferences and templates for documenting them with culturally contextual end-of-life care toolkits (e.g., spiritual rituals, last rites templates) • Distress protocols for managing families experiencing emotional distress heightened anxiety and anticipatory grief • Family leaflets and brochures on grief and bereavement support • Developing staff debrief cards for emotional fatigue and burnout tracking	• Creating a drug storage system for essential drugs for symptom management in ICU • Workshops and training ICU providers in symptom management, identifying family distress, eliciting after-death preferences and sensitively addressing the same • Setting up a bereavement care process through training hospital social workers or clinical psychologists • Institutionalize grief-response protocols for ICU nurses • Emergency psychological first aid SOPs for caregivers/families • Customizing bereavement leaflets based on religion, region, and language
7. Care Evaluation	• Guidance document on a bereavement phone call to families 3 weeks after the bereavement to know how the family is coping with the grief and collect informal feedback on care received at the end of life and their satisfaction • Manual on bereavement support to families, ensuring to include family feedback from poor, low-literacy, and rural voices • Manual and frameworks for debriefing and review of the care provided in the ICUs to improve care quality, utilizing feedback from both structured (FS-ICU/QODD) and open-ended feedback • Include metric for implementation climate and team burnout	• Incorporating bereavement phone calls and bereavement support as part of routine ICU care • Case reviews with patient/family voice included • Incorporating weekly debriefing and review of care-provided meetings for ICU deaths where end-of-life care was provided • Training ICU providers and hospital social work team/clinical psychologists in bereavement phone call and bereavement support • Training ICU providers in conducting debrief meetings and medical care review meetings.

		• Develop 'learning circle' modules for staff reflection and training refreshers • Establish independent care reviewers (outside primary ICU team) for quality audits – can be ICMR mentors
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Novelty

It is an evidence-based framework developed following an umbrella review(systematic review of systematic review) by the PI of this grant application published in Intensive Care Medicine Journal Q1-Top 1% IF 27 <https://link.springer.com/article/10.1007/s00134-024-07565-7> This framework is developed first-time worldwide through synthesis of 40 systematic reviews published worldwide and tested for the first time. Moreover, the PI of this grant application developed the guidelines for appropriate care in the ICU for the first time in 2019, revised in 2021. It can be accessed through this link <https://khmanipal.com/blue-maple/>. The ICMR, AIIMS Delhi and FICCI have referred to this framework while developing their own. This is an expansion of each component of the framework to know its implementation and impact across India. It integrates multidisciplinary inputs, such as ethical deliberation and family involvement, to enhance patient-centred care. It features a structured framework that balances clinical effectiveness, patient dignity, resource efficiency, and economic sustainability. Its adaptability spans various healthcare settings, from tertiary hospitals to community ICUs, and includes staff training, standardized decision-making protocols, and guidance on ethical dilemmas. It optimizes ICU resource allocation by applying health economic principles, promoting equitable and cost-effective care. This approach improves patient outcomes and aligns with global health initiatives, including several United Nations Sustainable Development Goals. Although some components of these intervention packages are implemented and tested in some ICU settings in India, the package as a total is never implemented, and its patient and family benefit in the Indian context is not known, making it a novel area of research enquiry.

Preliminary Work Done by the PI in the area of CAR Grant Application

1. In Association with ICMR, Developed Standard Definitions for Treatment Limitations and End-Of-Life Care for India, published on the ICMR website https://ethics.ncdirindia.org/asset/pdf/Definition_of_terms_used_in_limitation_of_treatment_and_providing_palliative_care_at_end_of_life.pdf. and published the same in Indian Journal of Critical Care Medicine¹.
2. Developed Guidelines for Withdrawal of Life Support in Terminally Ill Patients in association with the Ministry of Health Family Welfare <https://mohfw.gov.in/sites/default/files/Guidelines%20for%20withdrawal%20of%20Life%20Support.pdf>
3. Developed Ethics-Based Decision-making framework for Private Hospitals in India through the Federation of Indian Chamber of Commerce (FICCI) https://www.ficci.in/study_details/23119

4. Contributed to developing three policies and position statements on palliative care in Intensive Care Units published in the Indian Journal of Critical Care Medicine^{9, 10} and Indian Journal of Palliative Care¹⁸.
5. Developed India's first procedural guidelines for treatment limitation and end-of-life care in ICUs for India – A 50-page procedural guideline booklet titled BLUE MAPLE
<https://khmanipal.com/blue-maple/>
6. Set up India's first International Reference Centre for end-of-life Care in ICUs
<https://www.bestcareforthedying.org/reference-centres>
7. Worked with the NABH on developing accreditation provisions for End-of-Life Care in ICU (COP 20)
<https://nabh.co/wp-content/uploads/2024/01/Draft-NABH-Accreditation-standards-for-Hospitals-6th-Edition.pdf>
8. Developed India's first hospital-based guidelines for setting up a critical care review board (CCRB) and discharge against medical advice (DAMA) to enable patients in ICUs to receive appropriate care.
9. End of Life Care Provision in COVID ICUs, published in Indian Journal of Palliative Care¹⁹.

Proof of Concept Work to Support the CAR Grant Application

Preliminary Analysis – This brief report excludes 2024 data (Detailed Analysis Ongoing and including 2024 data the number of patients provided end-of-life care in ICUs is close to 2000)

Audit of implementing BLUE MAPLE (comprehensive palliative care package) in the Kasturba Medical College Hospital Manipal ICU setting. <https://khmanipal.com/blue-maple/>

BLUE MAPLE

Before Life Ends, Understand and Evaluate the Choice of Medical Treatment Offered
Methodised Action Plan for Limitation of Life-Sustaining Treatment and End of Life

BLUE MAPLE is an acronym for **B**efore **L**ife **E**nds, **U**nderstand and **E**valuate the Choice of Medical Treatment Offered: **M**ethodized **A**ction **P**lan for **L**imitation of Life-Sustaining Treatment and **E**nd of Life Care. It is a guidance document for initiating interventions towards limiting life-sustaining treatment and end of life care in patients with a critical and terminal illness.

Need for a BLUE MAPLE program

India does not have a national policy on end-of-life care, and the legal standpoint on limiting life-sustaining treatment is ambiguous. The lack of clarity has led to indiscriminate use of intensive care resources at the end of life, leading to higher ICU deaths, poorly managed symptoms, family distress and depletion of finances. The families often unilaterally initiate terminal discharges, and these discharged patients from the ICU do not receive any symptom relief or palliative care at the end of life. The BLUE MAPLE program provided a stepwise algorithm for acute care physicians to mitigate these circumstances.

Development and Implementation of the BLUE MAPLE program

It was developed by interdisciplinary experts from critical care, internal medicine, emergency medicine, paediatrics, neurology, neurosurgery, oncology and palliative care, hospital administrators, and a legal team. It was peer-reviewed by experts from India and internationally. We partnered with the International Collaborative for Best Care of the Dying Person, who conducted a 10/40 model congruence assessment and found it congruent. The 10/40 Model is derived from ten key principles and 40 goals of care that transcend international and cultural boundaries and apply irrespective of the place of care (**Supplement V**). Moreover, the end-of-life care section of BLUE MAPLE was adapted from the integrated care plan for the dying harmonized to the Indian context. The program was implemented in the ICU and acute care setting at Kasturba Hospital and associated hospitals of Manipal University.

Components of the BLUE MAPLE program

The BLUE MAPLE program has the following guidance documents and templates: a. recognition of medical futility in various clinical settings b. developing physician consensus on medical futility and template for endorsement of medical futility c. framework for communication and family meeting while discussing prognosis and template for documentation of family meeting d. consenting and template consent forms for withholding and withdrawing life-sustaining treatment e. ratification of futility, withholding/withdrawing decision by the hospital clinical ethics committee and template for ratification f. procedure for implementing withholding and withdrawing life-sustaining treatments and g. end-of-life care plan (initial and ongoing assessment and management), and after-death care protocol.

Study Design and Setting

Palliative care started integration with ICU services around 2018-2019. This retrospective observational study analysed adult patients admitted to the intensive care unit (ICU) from 2019 to 2023. Data were collected from the institutional database, excluding neonates and pediatric patients. The study aimed to evaluate ICU demographics, illness trajectories, palliative care (PC) referral patterns over the years, and end-of-life care documentation.

Study Population

A total of 1,417 adult patients (men: 940, women: 477) were included in the analysis. Data were stratified by demographics, payment systems, ICU types, and underlying critical illnesses necessitating ICU admission.

Data Collection and Variables

Data extraction focused on the following key variables:

1. Gender and age distribution.
2. Payment system utilized for ICU care.
3. ICU types and characteristics.
4. Underlying critical illnesses necessitating ICU admission (e.g., sepsis, stroke, cancer).
5. Preexisting comorbidities.
6. Illness trajectories are classified into four groups:
 - Acute onset with no premorbid illness.
 - Cancer-related ICU admission.
 - Chronic progressive organ impairment.
 - Frailty/neurodegenerative conditions

7. Length of stay and transfer dynamics:

- Time from hospital ward admission to ICU transfer.
- Direct admissions from the emergency department (ED) to the ICU.
- Time from ICU admission to palliative care referral.
- Duration from palliative care consultation to discharge or death.

8. Palliative care referral characteristics:

- Reasons for referral (e.g., symptom management, end-of-life care, family meetings).
- Documentation of care limitations, family meetings, and futility assessments.
- Outcomes of family meetings (e.g., treatment limitation, PC transfer, step-down care).
- Place of death (ICU, hospital ward, home, discharge against medical advice [DAMA]).

Descriptive statistics were used to summarize demographic and clinical characteristics. Temporal trends in palliative care referrals and documentation practices were analyzed across multiple years.

Results

Patient Demographics

Among 1,417 patients, 66.3% were male (n=940) and 33.7% were female (n=477).

Illness Trajectories and Length of Stay

The average number of days from ICU admission to palliative care referral was 6.8 days, with year-wise trends showing a decrease over time (2018: 6 days, 2019: 12 days, 2020: 9 days, 2021: 8 days, 2022: 7 days, 2023: 6 days). The breakdown of illness trajectories showed the following average durations:

- Acute illness: ICU stays 11 days.
- Cancer-related admissions: ICU stays 9 days.
- Chronic neurological diseases: ICU stays 11 days.
- Other chronic diseases: ICU stays 10 days.

ICU Admissions and Transfers

- The mean duration from ward admission to ICU transfer was 1.6 days.
- The proportion of direct ED-to-ICU admissions was unavailable.

Palliative Care Referral and End-of-Life Care

- In KMC Manipal, the critical care review board consists of ICU physicians, primary referral physicians, and PC physicians who meet every week to discuss cases admitted in the ICU when the LOS approaches a week to discuss the benefit of ICU continuation.
- Referrals to PC from the ICU saw a significant increase, with 36 cases in 2019, 121 in 2020, 215 in 2021, 448 in 2022, and 597 in 2023.
- The leading reasons for referrals to palliative care (PC) included symptom management (190 cases), family meetings (457 cases), and end-of-life care (785 cases).
- The average duration from ICU admission to PC referral was 6.8 days.
- Patients were seen by PC a mean of 3.9 days before discharge or death (median: 1 day, SD: 22.5 days).
- Documentation of end-of-life care increased over the years, with notable growth in family meetings and symptom management discussions

- The average interval between ICU admission and PC referral decreased from 12 days in 2019 to 6 days in 2023, reflecting improved referral efficiency. Similarly, the time from PC consultation to discharge averaged 4.2 days in 2019 and stabilized around 4 days in subsequent years. The average duration from ward admission to ICU peaked at 10 days in 2021 but decreased to 8 days in 2022 and 2023.

Documentation Trends

- The number of family meetings increased from 5 in 2018 to 182 in 2023.
- Documentation of treatment limitation and life-sustaining treatment (LST) withdrawal rose from 0 in 2018-2019 to 339 cases in 2023.
- Blue Maple futility documentation was reported in 258 cases in 2023.

Outcomes and Place of Death

- The proportion of patients discharged against medical advice (DAMA) declined from 19.44% in 2019 to 8.54% in 2023, indicating improved adherence to medical guidance and a shift from unhelpful ICU treatments.
- The outcomes of family meetings in an intensive care unit (ICU) setting:

Limitation of Life-Sustaining Treatment (796 cases) – This was the most common outcome, indicating a decision to withdraw or withhold life-prolonging medical interventions when deemed futile.

Non-Escalation of Care (793 cases) – A nearly equal number of cases involved the decision not to escalate treatment, ensuring that patients receive comfort-focused care without additional aggressive interventions.

Symptom Control (538 cases) – Many patients received symptom management strategies to improve their quality of life through palliative interventions.

De-Escalation of Care (443 cases) – This category reflects cases where ongoing interventions were scaled back gradually to prioritize comfort and reduce unnecessary medical procedures.

Terminal Sedation (439 cases) – Palliative sedation was provided for symptom relief in terminally ill patients to ensure comfort in their final moments.

Compassionate Extubation (389 cases) – Mechanical ventilation was withdrawn and controlled for a dignified and peaceful passing.

Finances: Healthcare expenses were primarily covered through out-of-pocket payments (722 cases), followed by public funding (407 cases) and private insurance (288 cases).

These findings highlight an evolving trend in ICU palliative care integration, increased documentation of care limitations, and enhanced communication regarding end-of-life care. Further research is warranted to explore patient-centred outcomes and optimize palliative interventions in critical care settings.

Learnings from Proof-of-Concept Work (Implementation of the package at the single ICU site)

What made implementing BLUE MAPLE feasible at Kasturba Medical College Hospital ICU?

- Open discussions between ICU providers, patients, and their families helped align treatment with patient/family preferences.
- Encouraging timely documentation of patient wishes, including do-not-resuscitate (DNR) orders and advance directives, prevented inappropriate treatments.

- Implementing evidence-based, comprehensive referral tools, including frailty as a criterion, SOFA scores, etc, and having a Critical Care Review Board helped identify palliative care needs in ICU patients.
- Conducting workshops to enhance ICU staff's knowledge in prognostication and empathetic communication improved palliative care integration.
- Promoting collaboration between ICU teams and palliative care through joint research projects ensured a coordinated approach, improved familiarity, and facilitated timely referrals.
- Support and the need for help from the hospital administration were crucial ingredients for the success of the BLUE MAPLE program's implementation.

What were some of the challenges in implementing BLUE MAPLE at Kasturba Medical College Hospital ICU?

- Limited understanding of palliative care scope and benefits by non-intensive care specialists managing patients in the ICU
- Perception that withholding or withdrawing life-sustaining treatment is illegal, despite legal protections.
- High rates of discharge against medical advice (DAMA) **are** due to DAMA's encouragement by the treating doctors.

Steps Taken to Implement BLUE MAPLE Program at Kasturba Medical College Hospital ICU

The BLUE MAPLE program was developed in alignment with joint society guidelines on end-of-life care (EOLC) set by the Indian Society of Critical Care Medicine and the Indian Association of Palliative Care. The initiative aimed to create a structured EOLC policy to guide appropriate medical interventions for patients with advanced life-limiting illnesses. A hospital end-of-life care task force comprising specialists from various medical disciplines, legal experts, and hospital administrators collaboratively developed the document. Following internal peer review, the document underwent a two-tier external peer review involving multidisciplinary clinicians and legal professionals across India. After revisions, the hospital quality team finalized and disseminated the document. The 12-step approach within the five-part document ensures ethical, patient-centred decision-making, improving the quality of end-of-life care.

Impact of Implementing BLUE MAPLE Program

A. In the last five years, the BLUE MAPLE was initiated in close to patients with terminal or critical illness in ICU/HDU setting, and the majority had limitations of treatment and received end of life care. B. The BLUE MAPLE in the first wave of the COVID-19 pandemic enabled 53 patients with serious COVID-19 illness and 75 patients in the second wave to receive end-of-life care. C. Audit of terminal discharge against medical advice has shown a progressive reduction of discharge against medical advice in the last two years. D. Shift in the attitudes of acute care physicians facilitating the integration of palliative care in the ICU setting. E. BLUE MAPLE has inspired two national bodies, ICMR and FICCI, to create guidance documents adapted from BLUE MAPLE for nationwide dissemination. F. Many hospitals across India have adopted this initiative in their hospital end-of-life care policy. G. Learnings from implementing BLUE MAPLE enabled us

to win a GCRF-UKRI pump-priming global grant in association with Lancaster University UK to further develop ICU palliative care in India.

It has led to the creation of India's First International Reference Centre for End-of-Life Care with a focus on ICU setting



Research Question

How can a Multifaceted, Comprehensive, Appropriate Care Package (MCACP) for people with advanced and terminal illnesses needing ICU support be co-developed, implemented, and iteratively refined across diverse Indian ICU contexts? & What is its impact on contextual adaptability, acceptability, implementation feasibility, and patient, family, and system-level outcomes such as quality of dying, family satisfaction, resource utilization, and cost?

Implementers, Implementation, and Evaluators (Harmonisation from International Evidence)

Based on a systematic review²⁰ conducted on Implementing palliative care in the intensive care unit, Implementers of interventions will be senior ICU nurses/nurse practitioners trained in palliative care and appropriate care interventions able to champion the cause through a role in supporting, marketing, and driving through an implementation, overcoming indifference or resistance that the intervention may provoke along with an interdisciplinary team coming in when needed on a consultative mode comprising of members of the multidisciplinary palliative care team or ICU providers trained in palliative care intervention.

We recognize that many centres are facing a shortage of manpower. To address this, we will provide additional staff for the duration of the research study. We hope that if the research proves successful and beneficial to these centres, they will consider funding these positions in the future. Our plan includes appointing a dedicated and trained ICU nurse practitioner as the key implementer (Project Nurse III) at each site. Additionally, we will hire two Project Research Scientists III—one focusing on medical aspects and the other on non-medical aspects—to oversee the research. We can also employ a social worker to help with family meetings. Another Non-Medical Research Scientist will assist with data analysis, both quantitative and qualitative, and will ensure quality control throughout the research process. A Senior Project Assistant will coordinate between the research staff and hospital team members and help in data analyses/management. Depending on the centre's needs, we can also appoint an Office Helper.

The package bundles will be implemented sequentially, following the natural progression of care: prognostication, communication, decision-making, care planning, care delivery, and care evaluation. However, these bundles can also be delivered iteratively to respond to any communication needs that arise after planned decisions or care. At the CAR centre, we will have additional manpower: Project Research Scientists III (Non-Medical) to work on the systematic review, health technology assessment (HTA), and the integration of research findings into public health and policy. Senior Project Assistants will help with project administration and financial management, while the Office Helpers will assist with office tasks and front desk research work. Moreover, we plan to appoint independent research assistants at each site to help mitigate reporting and confounding errors based on the study methods outlined below.

Mitigating reporting and confounding errors

The reporting errors will be mitigated through double-checking mechanisms, such as cross-referencing data with sources using predefined formats to minimize errors and educate those collecting data on proper reporting protocols. The confounding errors will be addressed by analysing data within subgroups to control for confounding variables, including potential confounders as covariates in statistical models to adjust for their effects, and testing different scenarios to assess the robustness of results in the presence of possible confounders. We will periodically organize external peer reviews of the project, utilizing ICMR and our mentors.

Objective 1: Co-develop the MCACP intervention through iterative, participatory design.

Design

- Participatory, mixed-methods approach using Delphi, Nominal Group Technique (NGT), and human-centred design (HCD).
- Iterative prototyping and usability testing of intervention components with frontline staff.
- Systematic reporting of MCACP using the TIDieR checklist. **(Supplement VI)**
- Stakeholder engagement through consensus-building workshops, utilizing explicit integration of power-sensitive and equity-focused co-design, accommodating institutional hierarchies, varied ICU capacities, and family perspectives.

Settings

- Stakeholders from diverse ICU settings across India (urban/rural, public/private, tertiary/district).
- Inclusion of institutions with and without existing palliative care integration.
- Deliberate engagement of under-resourced or low-readiness hospitals.

Sample Size

Delphi Process: ~50 stakeholders from multidisciplinary backgrounds.

Nominal Group Technique (NGT): 10–15 participants across roles, ensuring representation from nurses, social workers, and caregivers.

Usability Testing: 5–8 frontline ICU staff per pilot ICU site (for early-stage prototyping).

Participants

- ICU physicians (senior and junior), nurses (across cadres), and palliative care specialists.
- Hospital administrators, ethicists, legal experts.
- Caregiver/family representatives, especially from lower socioeconomic backgrounds. Types of caregivers we anticipate are - Primary family caregiver (spouse, child), Legal surrogate (next of kin), Professional caregiver (in institutions)
- Public hospital staff, district-level champions, and voices from tribal/rural communities.

Accrual, Outreach, and Contingency Measures

- Outreach via professional societies, patient networks, public sector health programs, and caregiving associations.
- Targeted inclusion of dissenting or underserved voices, with oversampling if needed.
- Recruitment via email invitations, professional meetings, and posters in clinical settings.
- Contingency: Extend recruitment timelines; leverage local site champions.

Resources Needed

- Human: Delphi/NGT facilitators with training in realist interviewing and HCD.
- Infrastructure: Virtual platforms, translation support, and field facilitators for inclusive sessions.
- Ethics support: Guidelines for handling distress-triggering conversations during co-design workshops.

Outcomes - Implementation outcomes

- Iteratively developed, field-validated, context-sensitive MCACP intervention.
- Early-stage acceptability and usability feedback from frontline users.

- Prototype refinement plan for downstream piloting and scaling.

Outcome Measures

- Completion of the TIDieR checklist for intervention reporting.
- Consensus on intervention components through Delphi rounds, and usability ratings across settings.
- Agreement scores among stakeholders in the Nominal Group Technique process.
- Reflexive documentation of power dynamics and implementation climate during workshops.

Project Implementation Plan

- Report the initial MCACP intervention developed from the synthesized MCACP¹¹ using the TIDieR checklist²¹
- Conduct Workshop 1: 3-hour workshop for Delphi participants on overview of appropriate care in ICUs, rationale for conducting this study, and components of MCACP intervention package and delivery
- Conduct 2-Step Delphi process using the Conducting and REporting of DELphi Studies (CREDES) approach²²
- Conduct Workshop 2: 3-hour workshop for nominal group technique (NGT) participants on overview of appropriate care in ICUs, the rationale for conducting this study, and components of MCAP intervention package and delivery co-designed from Delphi.
- Develop consensus for the co-designed MCAP using NGT²³
- Finalize the MCACP intervention package using the TIDieR checklist²¹
- Prepare a “minimum viable MCACP package” tailored for district/low-resource hospitals.

Duration

- 0-6 months

Objective 2: Pilot the MCACP intervention in two ICUs with existing palliative care integration to refine implementation and intervention and explore the key facilitators and barriers.

Design

- A mixed-methods, co-developmental study using the UK Medical Research Council (MRC) methodology for complex interventions (**Supplement VII**), integrated with the Consolidated Framework for Implementation Research (CFIR) and Normalization Process Theory (NPT) to guide process evaluation, stakeholder behaviour, and system integration.
- Designed to iteratively refine the MCACP model through real-world implementation feedback from diverse stakeholders in two ICU ecosystems.

Settings

KMC Manipal – Open ICU model, diverse specialty referrals, ICU consult-based.

St. John’s Medical College Hospital, Bangalore – Closed ICU model with full intensivist control.

Purposeful contrast to test intervention adaptability across different ICU governance and decision-making models.

Sample Size

- Observational: 50 participants/ per site
(10 ICU physicians, 10 non-ICU physicians, 10 ICU nurses, 10 family members, 5 administrative/public reps)
 - Qualitative: 25 participants
(5 ICU physicians, 5 non-ICU physicians, 5 nurses, 5 family members, 2 administrative, 3 public reps)
- Patients : we will recruit 100 patients per clinical site (**please see supplement VIII**)
- Optional: Include bereaved family feedback where ethically appropriate.

Participants

- ICU patients with terminal or advanced illnesses – detailed inclusion/exclusion criteria in **Supplement IX**.
- ICU and palliative care clinicians (including nurses).
- Families and caregivers, including representatives from under-represented groups.
- Public and organizational stakeholders (e.g., ethics committee, patient advocates).

Accrual, Outreach, and Contingency Measures

- Recruitment via palliative care referrals, family meetings, and ICU admission logs.
 - Diverse outreach channels: clinician networks, patient advocates, bereavement support groups.
- Contingency: Extend recruitment windows or adjust sampling across ICUs to meet diversity targets (e.g., socio-economic, gender, literacy).

Resources Needed

Human: Research staff trained in realist interviewing, NPT-guided facilitation, and trauma-informed care.

Infrastructural: Data platforms, multilingual interview aids, EHR access, simulation spaces for training.

Outcomes

Implementation outcomes

- Context-sensitive refinement of MCACP bundle elements.
- Identification of organizational readiness, cultural enablers/barriers, and real-world usability.
- Co-designed adaptations specific to closed vs. open ICU models.

Clinical outcomes: QODD, symptom relief like Opioid/sedative usage logs for symptom control, care de-escalation rates like - Number of patients receiving compassionate extubation, Rate of non-escalation decisions

Outcome Measures

- Fidelity and reach of each MCACP module (tracked via adapted TIDieR checklist).
- Staff training completion and retention rates.
- Qualitative themes on acceptability, relevance, and alignment with ICU workflows.
- Indicators of normalization (e.g., communication routines, shared decision-making uptake) per NPT domains

Project Implementation Plan

- Map stakeholder roles (developers, implementers, evaluators) and define relational accountability.
- Conduct contextual mapping using CFIR domains: readiness, leadership engagement, policy alignment.
- Use realist interviewing (NPT-informed) to understand behaviour shifts, tensions, adaptation.
- Run structured learning loops post-implementation (feedback sessions, debrief circles).
- Prototype updates refined collaboratively every 2 months.
- Document iterative refinements using the TIDieR checklist (**Supplement VI**) with logs of usability and ethical feedback.

Ethical and Training Considerations

Specialized training for qualitative interviewers in:

Responding to emotional distress.

Identifying and addressing emotional burden of ICU staff.

Realist and trauma-informed interviewing approaches.

Establish ethical escalation protocols for emotionally sensitive interviews.

Duration

6 months – 18 months, phased into:

- 0–3 months: Co-design adaptation
- 4–12 months: Pilot rollout and iterative learning
- 13–18 months: Consolidated refinements and readiness assessment

Objective 3: Expand implementation to four ICUs with existing palliative care access, with an aim of refining implementation and intervention, and evaluate its effectiveness on quality of dying, family satisfaction, cost, and resource utilization.

Design

- Implementation Science Lens:
Guided by CFIR for implementation design and barrier/facilitator mapping
Guided by NPT to evaluate normalization and embedding within ICU culture
- Research Phases:
Qualitative exploration of ICU ecosystem and stakeholder readiness (realist-informed interviews using NPT)
Mixed-methods implementation planning with real-world constraints and champions
Iterative implementation with embedded adaptation logic based on frontline feedback
Mixed-methods process evaluation (CFIR + NPT-aligned metrics)
Quasi-experimental pre-post outcome evaluation to assess real-world effectiveness

Settings

- Four hospital ICUs with palliative care access across India.

- **Inclusion Criteria:** Multidisciplinary ICU, with a minimum of 10 ICU beds, 3 Intensive Care Physicians, with 10 trained ICU nurses and willing to participate in the process of implementing MCACP interventions

Sample Size

1. A qualitative study to explore the ecosystem of implementation
25 participants (multi-stakeholders)
2. Mixed-methods study to plan implementation (observational + qualitative)
Observational – 80 participants (20 participants/ICU x 4 ICUs – multi-stakeholder)
Qualitative - 25 participants (multi-stakeholders)
3. Mixed-methods study to evaluate implementation (observational + qualitative)
Observational – 80 participants (20 participants/ICU x 4 ICUs – multi-stakeholder)
Qualitative - 25 participants (multi-stakeholders)
4. Quasi-experimental study (Pre-Post) to know the effectiveness of the intervention package
Based on the two published studies – one on the quality of dying in ICU²⁴ and an Indian study on family satisfaction in ICU²⁵, the sample sizes were 309 and 336 in two ICUs. Therefore, we would like a sample of 150/per ICU - 600 each pre- and post-phase.

Participants

- ICU and palliative care teams (doctors, nurses, social workers)
- Terminally ill ICU patients and their families
- Hospital administrators, policy influencers
- Public representatives (where possible)

Accrual, Outreach, and Contingency Measures

- Recruitment through participating hospital ICUs.
- Outreach via hospital networks, professional organizations, and patient groups.
- Contingency: Additional training sessions and adaptive implementation strategies. Training in qualitative methods, NPT, and implementation readiness. Site readiness assessed and iteratively supported

Resources Needed

- Human: Clinical teams, trained qualitative researchers, implementation leads, health economists
- Infrastructure: Implementation toolkits, SOPs, qualitative data platforms, telehealth support
- Training: On CFIR/NPT, qualitative realism, empathetic interviews, and adaptation workshops

Outcome

Implementation outcomes:

- Organizational readiness, implementation climate
- Staff adherence, fidelity, and contextual adaptation
- Embedding of MCACP in practice (NPT constructs: coherence, cognitive participation, collective action, reflexive monitoring)

Clinical outcomes: QODD, symptom relief like Opioid/sedative usage logs for symptom control, care de-escalation rates like - Number of patients receiving compassionate extubation, Rate of non-escalation decisions

Outcome Measures

- Quality of Dying: ICU-QODD scores; preference-concordant care delivery; symptom control
- Family Satisfaction: FS-ICU scores; interviews on communication, emotional support
- Care Cost: Direct + indirect ICU cost comparison; ICERs using EQ5D/QALYs; budget impact analysis
- Resource Utilization: Length of stay, procedures avoided, staff allocation/ adherence, PC referrals
- **Quality of Dying:** Scores on validated 25-item ICU-QODD and Percentage of patients whose end-of-life care aligns with documented preferences (e.g., advance directives, family-reported choices). Frequency of symptom management interventions (e.g., pain relief, opioid usage, sedation) tailored to patient comfort per the study protocol.
- **Family Satisfaction:** Scores on validated family satisfaction surveys (e.g., Family Satisfaction in the ICU, FS-ICU). Through qualitative interviews, explore family views on communication and involvement in decision-making processes. Exploring family experiences with care coordination and emotional support – through qualitative interviews.
- **Care Cost:** Cost of ICU care (direct and indirect cost) in patients with Advanced/Terminal illness with MCACP intervention + standard ICU care compared to standard care. Incremental cost-effectiveness ratio (ICER) of the intervention relative to improved outcomes (e.g., quality-adjusted life years, QALYs using EQ5D). $ICER = (Cost\ of\ Intervention - Cost\ of\ Standard\ Care) / (Effectiveness\ of\ Intervention - Effectiveness\ of\ Standard\ Care)$. ICER is expressed as cost per additional QALY gained (or other relevant outcome, such as improvement in quality of dying or family satisfaction). Cost savings associated with reduced ICU length of stay or avoidance of non-beneficial treatments. Budget impact analysis of scaling the intervention across multiple ICUs or healthcare systems. $Budget\ Impact = (Cost\ of\ Intervention - Cost\ of\ Standard\ Care) \times Number\ of\ Patients \times Adoption\ Rate$.
- **Resource utilization:** Average ICU length of stay for patients receiving the MCACP intervention compared to standard care. The number of invasive procedures or non-beneficial treatments avoided due to the intervention—utilization rates of palliative care services or hospice referrals. Staffing and resource allocation are required to sustain the intervention over time.

Project Implementation Plan:

- Conduct a qualitative study to explore the implementation ecosystem: Know the values, beliefs, attitudes, perceived barriers to implementation, and social, cultural, legal, political, technical, organizational, and funding factors in the implementation system that enable/constrain implementation. Public societal views. Conflicting/competing interests limiting implementation. Alignment of intervention with quality metrics, benchmarking and accreditation, enabling implementation. Do relevant participants make sense of the intervention, and does implementing MCACP align with their shared purpose? Do relevant participants feel they will benefit from implementing the interventions, and do they value them?

- Conduct a mixed-method study to do a pre-implementation assessment to plan implementation. Know the infrastructure needed for implementation (space, technology, staffing, resources). Relationships, communication and information-sharing practices between intervention implementers, participants and evaluators. Culture – Human equality, participant, recipient and learning-centredness. Perceived need for the intervention and change brought about by implementing intervention and intervention/change aligns with individual/organizational values. Resources are available for implementation (expertise, space, funding, know-how). Individuals championing the cause (champions have the authority, influence, expertise, motivation and skills to perform the role). Commitment and engagement of the participants
- Conduct a mixed-method study to evaluate the implementation process. Know the implementing team (who), intervention (what), implementing process (how), information about priorities, preferences, and needs (deliverers, recipients, participants), identify and appraise barriers and facilitators to implementation and delivery of the innovation, define goals and measures for implementation success, collect and discuss quantitative and qualitative information about the success of implementation and intervention, explore perception/views on implementation, intervention, and benefits of intervention, adaptation, modification, utility, relevance and applicability of intervention.
- Conduct a quasi-experimental study (Pre-Post) to know the effectiveness of the MCACP intervention package on the quality of dying, family satisfaction, care cost and resource utilization.

Duration

- 12 – 36 months (Including formative phase, implementation, and pre-post evaluation)

Objective 4: Extend implementation to four ICUs without existing palliative care access, with an aim of refining implementation and intervention, and evaluate its effectiveness on quality of dying, family satisfaction, cost, and resource utilization.

Methodology

- Same as Objective 3

Duration

- 24 – 48 months

Objective 5: Develop a Technical Resource Hub for Supporting ICUs Across India

Design

- A nationally scalable, modular knowledge translation and implementation support platform developed through iterative co-design with stakeholders from diverse ICU contexts.
- The platform will support implementation, adaptation, capacity building, and equity-focused innovation in end-of-life ICU care.

Settings

- Nationwide ICU and palliative care networks - Government and private ICU networks, including district hospitals, teaching hospitals, and palliative care-integrated centres, Pilot engagement with low-resource ICUs to validate the relevance of resource hub content.

Sample Size

- To be determined based on the volume of content and diversity of settings – we will try to include testing educational materials and other resources with at least one district hospital and one tertiary centre to validate relevance across resource gradients.

Participants

- ICU and palliative care providers (doctors, nurses, AYUSH practitioners).
- Hospital administrators, quality and ethics committee members, and public health system officials.
- Policymakers (MOHFW, NHA, DGHS) and patient/family representatives, especially those from vulnerable communities.

Accrual, Outreach, and Contingency Measures

- Nationwide outreach through ICMR, FICCI, NABH, and palliative care networks.
- SWAYAM-like e-learning courses to engage clinicians and nurses at scale.
- Optional training credit pathways for ICU staff to encourage uptake.
- Contingency: Collaborate with state-level medical education departments and district hospital directors for sustained reach.

Resources Needed

Human: Content developers, ICU and palliative experts, IT developers, language translators, ethics advisors, and instructional designers.

Infrastructural: Secure digital repository, multilingual hosting infrastructure, mobile-accessible modules, and offline-compatible content kits (for low-connectivity regions).

Outcomes – Implementation and public health outcomes:

- Establishment of a centralized, public-facing resource and learning hub containing protocols, templates, training modules, and FAQs that facilitate:
 - Appropriate ICU care delivery.
 - Website/ E-learning for staff and awareness for members of the public.
 - We will offer offline training kits (PDFs, videos on USB drives, printed SOPs), and Translate resources into regional languages to increase reach.
- Policy advocacy and standardization across institutions - A "Minimum Viable MCACP Package" designed for low-resource settings will be co-created. The proposal will differentiate between core and optional components, particularly for low-resource environments.

Outcome Measures

- Number of hospitals using the MCACP tools from the hub.
- Uptake and completion of training modules (web-based or offline).
- Download/print metrics of SOPs and care planning templates.
- Number of regional adaptations and translated versions created.

- Visibility impact: Number of public awareness materials used in community sessions/media.

Project Implementation Plan

- Co-develop and Peer-Review Resources

Manuals, videos, infographics, SOPs, case studies, patient decision aids.

Ensure representation of voices from district hospitals, families, and nurses.

External peer review with ethics and public health stakeholders.

- Platform Design & Hosting

Select platform with open-access capability, intuitive navigation, and language toggle.

Design for scalability: module-based structure (e.g., Prognostication, Communication, Bereavement).

- Content Dissemination and Training

Upload and roll out training resources.

Conduct orientation for hospital teams via webinars and onsite sessions.

Pilot blended learning programs in 2–3 test sites.

- Policy and Public Integration

Align platform outputs with NABH, NHM, NHA benchmarks.

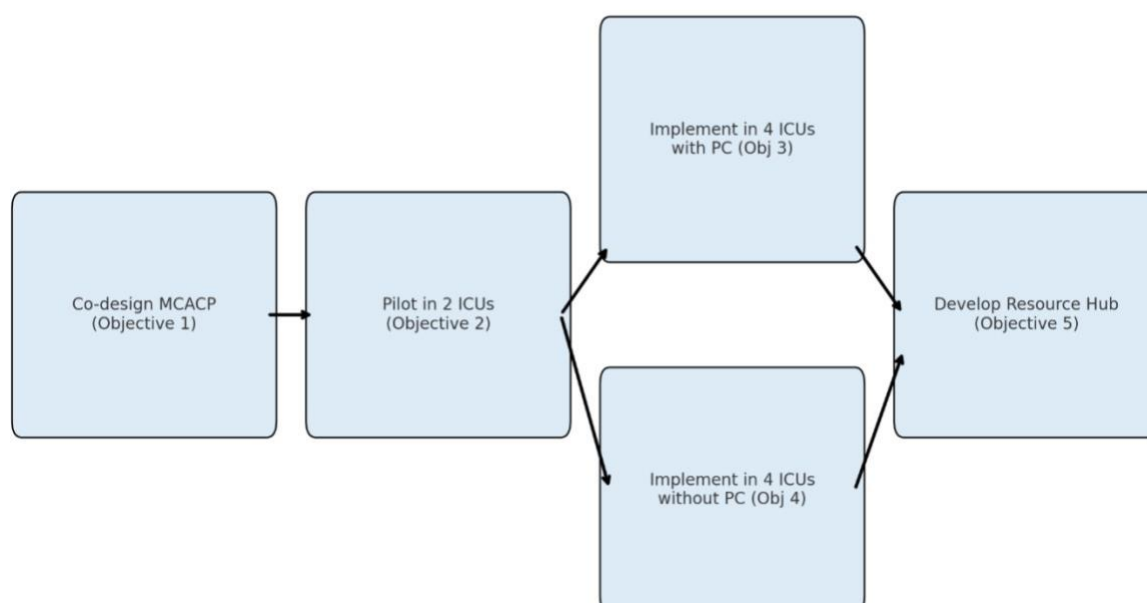
Enable health system actors to report implementation outcomes through the hub.

- Maintenance and Handover

Develop a roadmap to transfer hosting and maintenance to ICMR/DGHS.

Build capacity for periodic content updates and multilingual expansion.

Flow Diagram: Implementation of MCACP Across Objectives



Models for Intervention and Implementation

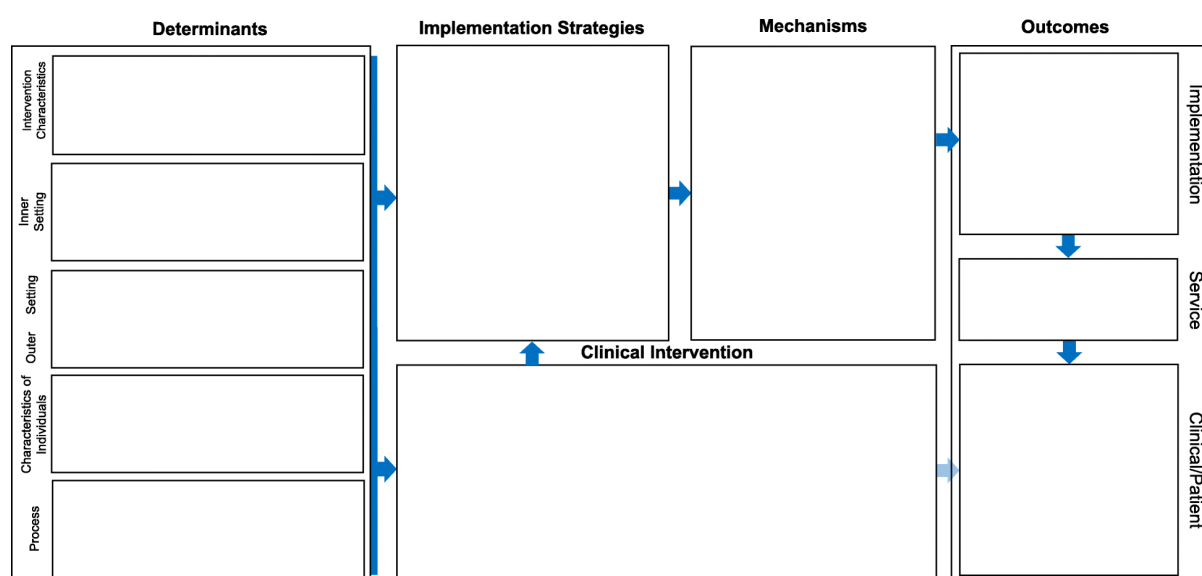
A three-tiered model for intervention and implementation will be developed

Model 1: Intervention and Implementation Framework for ICUs with Palliative Care Integration

Model 2: Intervention and Implementation Framework for ICUs with Palliative Care Access

Model 3: Intervention and Implementation Framework for ICUs with no Palliative Care Access

The models will be developed using the Implementation Research Logic Model (IRLM)



Determinants of implementation are factors that might prevent or enable implementation

Implementation strategies are supports, changes to, and interventions on the system to increase adoption of evidence-based interventions into usual care

Mechanisms of action are processes or events through which an implementation strategy operates to affect desired implementation outcomes

Implementation outcomes are the effects of deliberate and purposive actions to implement new treatments, practices, and services

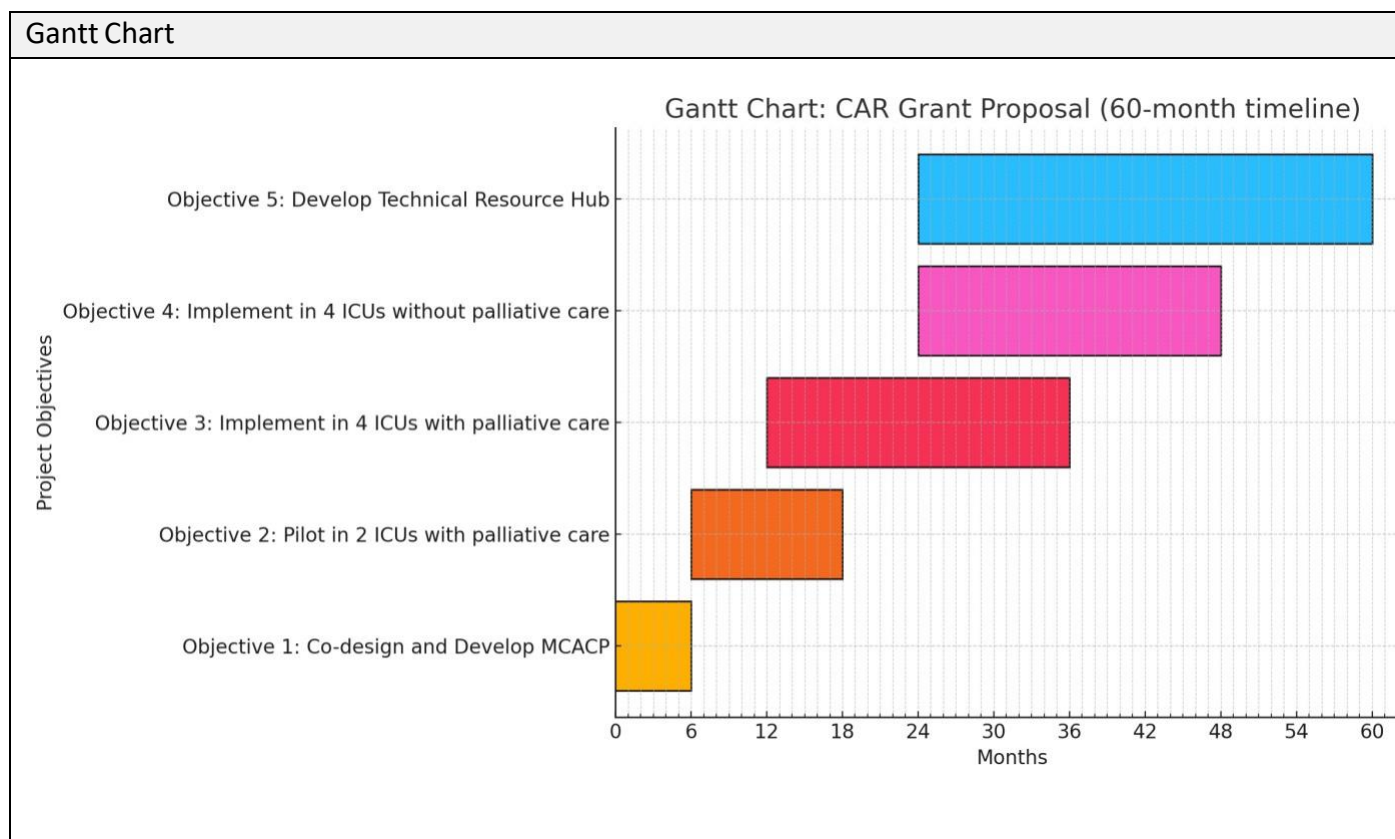
Table 2: Steps involved in creating resources for ICU providers to implement appropriate care package

Step 1 Content Organization	1.1. Creating Content Repository (generated during WP2, WP3, and WP4) – texts, videos, illustrations, guidance documents, tools, etc. 1.2. Content Categorization and Making a Manual 1.3. External peer-review of the content and the manual
Step 2 A platform for Hosting the Content	2.1. Choosing a suitable platform based on accessibility, scalability, search capabilities, and integration with existing systems 2.2. Creating a user-friendly layout with intuitive navigation, precise search functions
Step 3 Content Delivery	3.1. Uploading content into the platform, including text, images, videos, and interactive elements 3.2. Training and communication about the resources
Step 4 Content Maintenance	4.1. Handover the content and the content platform to ICMR/DGHS for regular updates of the content 4.2. Provide ongoing technical support to ICMR/DGHS for content updates and maintenance
Step 5 Content Utilisation	5.1. Develop strategies and metrics for uptake of the MCACP intervention package and plan to implement the MCACP package across India. 5.2. Develop strategies and metrics to measure outcomes of the MCACP intervention package across India.

Table 3. Consolidated Framework for Implementation Research (CFIR)-Normalization Process Theory (NPT) Integration

Framework Component	Application in the Study	Corresponding Methods/Tools
CFIR: Intervention Characteristics	Assess the adaptability and complexity of the MCACP across diverse ICU settings	Delphi consensus, TIDieR checklist, pilot refinement
CFIR: Outer Setting	Evaluate patient/family needs, societal views, and external policy/legal influences	Public representative interviews, socio-legal context mapping
CFIR: Inner Setting	Examine ICU culture, leadership engagement, organizational readiness	Qualitative interviews with ICU staff and administrators
CFIR: Characteristics of Individuals	Understand the knowledge, beliefs, and self-efficacy of ICU providers in delivering MCACP	Pre-post staff surveys, focused IDIs
CFIR: Implementation Process	Guide planning, engaging, executing, and reflecting during implementation phases	Project logs, field observations, weekly review meetings
NPT: Coherence	Explore how staff understand the purpose and meaning of MCACP	Interview probes on perceived value and purpose of intervention

NPT: Cognitive Participation	Identify engagement levels and belief in the intervention's legitimacy	Interviews with staff champions and resisters
NPT: Collective Action	Investigate operational integration into daily workflow	Observation of clinical rounds, communication documentation audits
NPT: Reflexive Monitoring	Assess how staff appraise and modify their use of MCACP over time	Post-implementation feedback sessions debrief meetings



Clinical Sites Identified for this Study		
Objective 1		
Kasturba Medical College Hospital Manipal		
Objective 2	Objective 3	Objective 4
- Kasturba Medical College Hospital Manipal - St John's Medical College Hospital Bangalore	- AIIMS Delhi - CMC Vellore - Tata Memorial Hospital Mumbai - Bharathi Vidyapeeth Hospital Pune	- Vardhaman Mahavir Medical College Safdarjung Hospital Delhi - Aster CMI Hospital Bengaluru - Yashodha Hospital Ghaziabad - P D Hinduja Hospital Mumbai

Technical Sites Identified for this Study

1. PGIMER Chandigarh

- Participates in health economic evaluation and integration of guidelines in health policy – objective 5

2. ICMR Bioethics Unit, Bengaluru

- Ethical oversight and peer review of intervention components (manuals, toolkits, training modules) – objectives 2 - 5.
- Ensures co-designed materials comply with national bioethics standards – objectives 2 - 5.
- Guides framing of consent protocols and sensitive topics like bereavement and treatment limitation – objectives 2 - 5.

3. Ministry of Health & Family Welfare (MoHFW), Delhi

- Policy integration and advocacy aligns outputs of the resource hub with national healthcare initiatives (e.g., NHM, NHA benchmarks) – objective 5.
- Provides long-term institutional ownership: the content platform is designed to be transferred and maintained by ICMR/DGHS/MoHFW – objective 5.
- Aids in dissemination and national rollout through central authority channels – objective 5.

4. Vidhi Centre for Legal Policy, Delhi

- Supports legal review and development of:
- End-of-life decision-making frameworks - objectives 2 - 5.
- Institutional policies regarding treatment limitation and patient rights – objective 5.
- Ensures MCACP practices align with legal feasibility in Indian ICUs and are grounded in clear constitutional and health law interpretations - objectives 1 - 5.

5. Karunashraya Institute for Palliative Care Education and Research (KIPCER), Bengaluru

- Contributes to the design of palliative care training content for ICU clinicians - objectives 1 - 5.
- Leads or co-leads modules on:
 - Communication during bereavement - objectives 2 - 5.
 - Staff debriefings and managing burnout - objectives 2 - 5.
 - Part of the resource hub creation to develop and host education material and modules - objectives 2 - 5.

6. Prasanna School of Public Health, MAHE, Manipal

- Participates in health economic analysis, outcome mapping, and public health translation - objectives 2 - 5.
- Guides strategies for:
 - Cost-effectiveness and resource use - objectives 2 - 5.
 - Frameworks for scalability in district and low-resource hospitals - objectives 2 - 5.
 - Co-leads on designing measurable outcomes, uptake indicators, and utilization metrics - objectives 2 - 5.

Expected Societal Outcomes and Impact

- Enhanced prognostic clarity enables more personalized and informed care.
- Improved communication empowers healthcare providers to navigate challenging conversations and fosters rapport between families and clinicians.
- Decision-making frameworks streamline care planning, reducing delays in care delivery and ensuring care alignment with patient values and clinical realities.
- Care planning honours patient and family preferences, promoting dignity and autonomy in caregiving.
- Care delivery frameworks aim to enhance the quality of life and death for patients who have had ICU care withheld or withdrawn while also providing psychosocial support for their families.
- Cost-effective implementation strategies are designed to facilitate the scalability and sustainability of intervention, enabling widespread adoption across diverse healthcare settings in India.
- This intervention is poised to establish a replicable model that aligns ethical care with improved health economics, ultimately redefining ICU care standards in India and other LMICs.

Potential Stratified Recommendations through Outcomes of this CAR Grant Proposal to India**Individual Level (Healthcare Providers and Patients)**

- Train oncologists and ICU clinicians to understand illness trajectory, disease outcomes, and palliative care needs
- Establish clear referral criteria for palliative care between ICU and palliative care providers
- Educate families on disease course, treatment options, symptom management, and the dying phase
- Shift from a purely biomedical model to a bio-psycho-social-spiritual approach

Organizational Level (Hospitals & Healthcare Institutions)

- Develop multidisciplinary ICU and palliative care teams.
- Implement pathways to help patients and families navigate healthcare systems.
- Train front-line healthcare workers in the ICU in primary holistic palliative care.
- Establish systems to ensure uninterrupted supply of essential medications for symptom management

National Level (Healthcare Systems & Policies) - sustainable, equitable, and resource-aware package for India

- Activate health coverage for palliative care needs in ICU through government schemes and private insurance.
- Enhance generalist palliative care service delivery and integrate it into ICU workflows.
- Mandate palliative care inclusion in all acute care training.
- Develop culturally specific psychosocial and care guidelines contextual to Indian ICUs
- Develop a central data collation portal for ICU PC for multicentre integration and real-time analysis.

Public Level (Community & Society)

- Establish community support systems to assist families with caregiving caring for patients in ICUs.
- Address stigma, myths, and taboos surrounding palliative care, death, and opioid use.
- Use family testimonials to increase public awareness and acceptance of palliative care in ICUs.

- Raise awareness about the symptom burden and suffering and the need to prioritize appropriate care.

Policy Makers (Government & Regulatory Bodies)

- Develop and enforce policies for limiting inappropriate life-sustaining treatment and end of life care in ICUs.
- Ensure funding for palliative care programs and packages in ICU through the national health authority
- Build partnerships to exchange knowledge on palliative care development in ICUs through national education initiatives.
- Engage opinion leaders and policymakers to drive advocacy and awareness about appropriate care in ICUs.

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List of supplements:**Supplement I:****Emergency Toolbox for Symptom Management of ICU Patients with Advanced Illness****Emergency Resources**

Personal protection equipment guidance, aerosol and airborne infection protection guide, hand hygiene and infection control protocol, pain scale, dyspnoea scale, delirium screening scale, symptom management algorithms, standard templates for a family meeting, documenting patient and family communication and foregoing life-sustaining treatments.

Emergency Medications

Morphine (Oral Immediate Release Tablet 10mg and Injection 10mg/ml), Midazolam (Injection 5mg/5ml), Lorazepam (Tablet 0.5 mg and 1mg), Metoclopramide (Tablet 10 mg and Injection 5mg/ml), Haloperidol (Tablet 0.5mg, 5 mg and Injection 5mg/ml), Paracetamol (Tablet 500mg and 650 mg and Injection 1gm/100ml), Pantoprazole (Tablet 40mg and Injection 40mg/vial), Dexamethasone (Tablet 4mg and Injection 4mg/ml), Hydrocortisone (Injection 100mg/vial), Furosemide (Injection 40mg/4ml), Amitriptyline (Tablet 10mg and 25mg), Citalopram (Tablet 10mg), Quetiapine (Tablet 25mg), Olanzapine (Tablet 2.5mg and 5mg), Glycopyrrolate (Injection 0.2mg/ml), Linctus Codeine Suspension, Phenobarbitone (Injection 200mg/ml) and Naloxone (Injection 0.4mg/ml)

Emergency Equipment

Personal protective equipment, opioid locked boxes, nasogastric tubes, urinary catheters, wound dressing, suction apparatus, portable oxygen, subcutaneous and intravenous catheters and lines, and syringe drivers if available.

Symptom management of patients withheld/withdrawn from the ventilator

Patients who are not ventilated or have discontinued ventilation can develop severe breathlessness, delirium, and moist breathing. They can be considered a combination of continuous infusion, intermittent dosing, and breakthrough medications.

Inj Morphine 10-15 mg/24h + Inj Midazolam 10-15 mg/24h can be combined and administered as an infusion or Inj Morphine 2mg + Inj Midazolam 2mg every 4 hours. If respiratory secretions are present, Inj Glycopyrrolate is 0.2 mg every 6-8 hours.

Breakthrough medications are given SOS for symptoms, and breakthrough medications can be given at one to two hours as required. The breakthrough doses are one-sixth of the 24-hour dose

Anticipatory Prescribing

Patients with serious illnesses not ventilated/withdrawn from a ventilator can develop severe symptoms. These should be anticipated, and an anticipatory prescription should be provided for all the patients.

Anticipatory prescription for patients with serious illnesses not ventilated or withdrawn from ventilator

Symptom Anticipated	Treatment Plan
Pain	Inj Morphine 1-2 mg SC or IV
Breathlessness	Inj Morphine 1-2 mg SC or IV
Distress/Agitation	Inj Midazolam 1-2mg SC or IV
Delirium	Inj Haloperidol 1-2 mg SC or IV
Delirium with severe agitation	Inj Haloperidol 1-2 mg SC or IV + Inj Midazolam 1-2mg SC or IV

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Respiratory secretions	Inj Glycopyrrolate 0.2 mg SC or IV
Nausea and Vomiting	Inj Metoclopramide 20mg SC or IV

Supplement II:

Quality of Dying and Death (QODD) — is an instrument for assessing the quality of dying and death. This version is designed for self-administration by a significant other (family member and/or friend) of a patient who died in a hospital or ICU setting.

During the last several days that your loved one was in the ICU:

1a. How often did your loved one appear to have his/her pain under control?

(Circle one number)

- 0 None of the time
- 1 A little bit of the time
- 2 Some of the time
- 3 A good bit of the time
- 4 Most of the time
- 5 All of the time
- 6 Don't know >>>>>>>> Go to Question 2a.

b. How would you rate this aspect of your loved one's dying experience?

(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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2a. How often did your loved one appear to have control over what was going on around him/her?

(Circle one number)

- 0 None of the time
- 1 A little bit of the time
- 2 Some of the time
- 3 A good bit of the time
- 4 Most of the time
- 5 All of the time
- 6 Don't know >>>>>>>> Go to Question 3a.

b. How would you rate this aspect of your loved one's dying experience?

(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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During the last several days that your loved one was in the ICU:

3a. How often was your loved one able to feed her/himself? (Circle one number)

- 0 None of the time
- 1 A little bit of the time
- 2 Some of the time
- 3 A good bit of the time
- 4 Most of the time
- 5 All of the time
- 6 Don't know >>>>>>>> Go to Question 4a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfec t
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4a. How often did your loved one appear to breathe comfortably?
(Circle one number)

- 0 None of the time
- 1 A little bit of the time
- 2 Some of the time
- 3 A good bit of the time
- 4 Most of the time
- 5 All of the time
- 6 Don't know >>>>>>>> Go to Question 5a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfec t
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During the last several days that your loved one was in the ICU:

5a. How often did your loved one appear to feel at peace with dying?
(Circle one number)

- 0 None of the time
- 1 A little bit of the time
- 2 Some of the time
- 3 A good bit of the time
- 4 Most of the time
- 5 All of the time
- 6 Don't know >>>>>>>> Go to Question 6a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfec t
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6a. How often did your loved one appear to be unafraid of dying? (Circle one number)

- 0 None of the time
- 1 A little bit of the time
- 2 Some of the time
- 3 A good bit of the time
- 4 Most of the time
- 5 All of the time
- 6 Don't know >>>>>>>> Go to Question 7a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfec t
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During the last several days that your loved one was in the ICU:

7a. How often did your loved one laugh and smile? (Circle one number)

- 0 None of the time
- 1 A little bit of the time
- 2 Some of the time
- 3 A good bit of the time
- 4 Most of the time
- 5 All of the time
- 6 Don't know >>>>>>>> Go to Question 8a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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8a. How often did your loved one appear to keep his/her dignity and self-respect?
(Circle one number)

- 0 None of the time
- 1 A little bit of the time
- 2 Some of the time
- 3 A good bit of the time
- 4 Most of the time
- 5 All of the time
- 6 Don't know >>>>>>>> Go to Question 9a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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During the last several days that your loved one was in the ICU:

9a. How often did your loved one spend time with his/her family or friends?
(Circle one number)

- 0 None of the time
- 1 A little bit of the time
- 2 Some of the time
- 3 A good bit of the time
- 4 Most of the time
- 5 All of the time
- 6 Don't know >>>>>>>> Go to Question 10a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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10a. How often did your loved one spend time alone? (Circle one number)

- 0 None of the time
- 1 A little bit of the time
- 2 Some of the time
- 3 A good bit of the time
- 4 Most of the time
- 5 All of the time
- 6 Don't know >>>>>>>> Go to Question 11a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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The following questions are answered with either a "Yes" or "No" based on whether your loved one did certain activities. Please rate the quality of that aspect of the dying experience. Again, we are asking you to focus on your loved one's last several days.

During the last several days that your loved one was in the ICU:

11a. Was your loved one touched or hugged by his/her loved ones? (Circle one number)

- 1 Yes
2 No
3 Don't know >>>>>>>> Go to Question 12a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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12a. Were all of your loved one's health care costs taken care of? (Circle one number)

- 1 Yes
2 No
3 Don't know >>>>>>>> Go to Question 13a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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REMEMBER: IF YOU HAVE ANY QUESTIONS, PLEASE CALL.

During the last several days that your loved one was in the ICU:

13a. Did your loved one say goodbye to loved ones? (Circle one number)

- 1 Yes
2 No
3 Don't know >>>>>>>> Go to Question 14a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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14a. Did your loved one clear up any bad feelings with others? (Circle one number)

- 1 Yes
2 No
3 Don't know >>>>>>>> Go to Question 15a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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15a. Did your loved one have one or more visits from a religious or spiritual advisor?
(Circle one number)

- 1 Yes
2 No
3 Don't know >>>>>>>> Go to Question 16a.

b. How would you rate this aspect of your loved one's dying experience?
(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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During the last several days that your loved one was in the ICU:

16a. Did your loved one have a spiritual service or ceremony before his/her death?

(Circle one number)

- 1 Yes
- 2 No
- 3 Don't know >>>>>>>> Go to Question 17a.

b. How would you rate this aspect of your loved one's dying experience?

(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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17a. Did your loved one receive a mechanical ventilator (respirator) to breathe for him/her? (Circle one number)

- 1 Yes
- 2 No
- 3 Don't know >>>>>>>> Go to Question 18a.

b. How would you rate this aspect of your loved one's dying experience?

(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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18a. Did your loved one receive dialysis for his/her kidneys? (Circle one number)

- 1 Yes
- 2 No
- 3 Don't know >>>>>>>> Go to Question 19a.

b. How would you rate this aspect of your loved one's dying experience?

(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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Please answer either "Yes" or "No" if your loved one ever experienced the following. Then, rate the quality of this aspect of your loved one's dying experience.

19a. Did your loved one have his or her funeral arrangements in order prior to death? (Circle one number)

- 1 Yes
- 2 No
- 3 Don't know >>>>>>>> Go to Question 20a.

b. How would you rate this aspect of your loved one's dying experience?

(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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20a. Did your loved one discuss his or her wishes for end of life care with his/her doctor -- for example, resuscitation or intensive care? (Circle one number)

- 1 Yes
- 2 No
- 3 Don't know >>>>>>>> Go to Question 21a.

b. How would you rate this aspect of your loved one's dying experience?

(Circle one number)

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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21a. Was anyone present at the moment of your loved one's death? *(Circle one number)*

- 1 Yes
- 2 No
- 3 Don't know >>>>>>>> Go to Question 22a.

b. How would you rate this aspect of your loved one's death? *(Circle one number)*

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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22a. In the moment before your loved one's death, was he/she: *(Circle one number)*

- 1 Awake
- 2 Asleep
- 3 In a coma or unconscious
- 4 Don't know >>>>>>>> Go to Question 23.

b. How would you rate this aspect of your loved one's death? *(Circle one number)*

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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23. Overall, how would you rate the quality of your loved one's dying? *(Circle one number)*

Terrible	0	1	2	3	4	5	6	7	8	9	10	Almost Perfect
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24. Rate the care your loved one received from all doctors and other health care providers (including nurses, caseworkers, and other health care professionals) during the last several days of his or her life while in the ICU. *(Circle the number)*

Worst Healthcare Possible	0	1	2	3	4	5	6	7	8	9	10	Best Healthcare Possible
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25. Rate the care your loved one received from his or her doctor during the last several days of his or her life while in the ICU. *(Circle the number)*

Worst Healthcare Possible	0	1	2	3	4	5	6	7	8	9	10	Best Healthcare Possible
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REMEMBER: IF YOU HAVE ANY QUESTIONS, PLEASE CALL.

ABOUT YOU

In this section, we would like to ask a few questions about you and about your loved one.

1. **When were you born?** *(Please write the year)*

19 _____

2. **When was your loved one born?** *(Please write the year)*

19 _____

3. **What is your gender?** *(Circle one number)*

- 1 Male
2 Female

4. **What is your loved one's gender?** *(Circle one number)*

- 1 Male
2 Female

5. **Approximately how many days was your loved one in the hospital?**
(Please write the number of days)

_____ days

6. **Approximately how many days was your loved one in the intensive care unit (ICU)?** *(Please write the number of days)*

_____ days

7. **What is your ethnicity?** *(Circle one number)*

- 1 Hispanic
2 Non-Hispanic

8. **What is your race?** *(Circle all that apply)*

- 1 White
2 Black / African American
3 Asian
4 Pacific Islander
5 Native American or Alaskan Native
6 Other (please specify) _____

9. **What is the highest level of schooling you have completed?** *(Circle one number)*

- 1 8th grade or less
2 Some high school
3 High school diploma or GED
4 Some college or trade school
5 4-year college degree (e.g. BA, BS)
6 Graduate or professional school

10. **How are you related to your loved one?** *(Circle one number)*

- 1 I am his/her spouse or partner
2 I am his/her child
3 I am his/her sibling
4 I am his/her parent
5 I am another relative
6 I am his/her friend
7 Other (please specify) _____

11. **Did you live with your loved one?** *(Circle one number)*

- 1 Yes
2 No

12. How long have you known your loved one? (Please fill in)

_____ number of years OR _____ number of months

13. Today's date is: (Please fill in today's date)

____ / ____ / 20____
Month Day Year

14. We would like to get feedback from you on how burdensome it was to complete this questionnaire. This information will help guide us in future research. Overall, how much of a burden on you was this questionnaire? (Circle one number)

No burden at all	Moderate burden										Great burden	
	0	1	2	3	4	5	6	7	8	9	10	

Thank you for taking the time to complete this survey. If you have any comments, please feel free to add them to the margins of the survey or to the space below, or call to talk with study staff. Thank you again for your help.

Adapted from -

*Glavan BJ, Engelberg RA, Downey L, Curtis JR. Using the medical record to evaluate the quality of end-of-life care in the intensive care unit. Crit Care Med. 2008 Apr;36(4):1138-46.

*Curtis JR, Downey L, Engelberg RA. The quality of dying and death: is it ready for use as an outcome measure? Chest. 2013 Feb 1;143(2):289-91. doi: 10.1378/chest.12-1941.

Supplement III:**Family Satisfaction with Care in the Intensive Care Unit: FS-ICU 24R©*****How are we doing?***

We would like to hear your opinions about your family member's recent admission to the Intensive Care Unit (ICU)

Your family member was a patient in this ICU. The questions that follow ask **YOU** about your family member's **most recent ICU admission**. We understand that there were probably many doctors and nurses and other staff involved in caring for your family member. We know that there may be exceptions but we are interested in **your overall assessment** of the quality of care we delivered. We understand that this was probably a very difficult time for you and your family members. We would appreciate you taking the time to provide us with your opinion. Please take a moment to tell us what we did well and what we can do to make our ICU better. **Please be assured that all responses are confidential.** The Doctors and Nurses who looked after your family member will not be able to identify your responses.

PART 1: SATISFACTION WITH CARE

Please check one box that best reflects your views. If the question does not apply to your family member's stay then check the 'Not Applicable' box (N/A).

HOW DID WE TREAT YOUR FAMILY MEMBER (THE PATIENT)? HOW SATISFIED ARE YOU WITH...

- | | | | | | | |
|---|---|---|---|---|---|--------------------------|
| 1. Concern and Caring by ICU Staff?
The courtesy, respect and compassion your family member (the patient) was given. |  |  |  |  |  | N/A |
| | Very Dissatisfied | Slightly Dissatisfied | Mostly Satisfied | Very Satisfied | Completely Satisfied | ☐ |
| 2. Symptom Management? | | | | | ☐ 5 | |

How well the ICU staff assessed and treated **your family member's** symptoms.

- | | | | | | | |
|---|---|---|---|---|---|----------------------------|
| a. How well the ICU staff assessed and treated your family member's <u>pain</u> . |  |  |  |  |  | N/A |
| | Very Dissatisfied | Slightly Dissatisfied | Mostly Satisfied | Very Satisfied | Completely Satisfied | ☐ |
| b. How well the ICU staff assessed and treated your family member's <u>breathlessness</u> . |  |  |  |  |  | N/A |
| | Very Dissatisfied | Slightly Dissatisfied | Mostly Satisfied | Very Satisfied | Completely Satisfied | ☐ 5 |



c. How well the ICU staff assessed and treated your family member's agitation.	Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
	☐	☐	☐	☐	☐	☐

HOW DID WE TREAT YOU? HOW SATISFIED ARE YOU WITH...

3. Consideration of your needs? How well the ICU staff showed an interest in your needs.						N/A
	Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
	☐	☐	☐	☐	☐	☐
4. Emotional support? How well the ICU staff provided emotional support to you .						N/A
	Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
	☐	☐	☐	☐	☐	☐
5. Co-ordination of care? The teamwork of all the ICU staff that took care of your family member.						N/A
	Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
	☐	☐	☐	☐	☐	☐
6. Concern and Caring by ICU Staff? The courtesy, respect and compassion you were given.						N/A
	Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
	☐	☐	☐	☐	☐	☐

THE ICU STAFF: HOW SATISFIED ARE YOU WITH...

7. Skill and Competence of ICU Nurses? How well the nurses cared for your family member.						N/A
	Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
	☐	☐	☐	☐	☐	☐
8. Frequency of Communication With ICU Nurses? How often nurses communicated to you about your family member's condition.						N/A
	Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
	☐	☐	☐	☐	☐	☐

9. **Skill and Competence of ICU Doctors (All Doctors including Residents)?**
How well doctors cared for your family member.
- | | | | | | |
|---|---|---|---|---|--------------------------|
|  |  |  |  |  | N/A |
| Very Dissatisfied | Slightly Dissatisfied | Mostly Satisfied | Very Satisfied | Completely Satisfied | ☐ |
| ☐ 1 | ☐ 2 | ☐ 3 | ☐ 4 | ☐ 5 | |

THE WAITING ROOM

10. How satisfied are you with the atmosphere (mood) in the ICU waiting room?
- | | | | | | |
|---|---|---|---|---|--------------------------|
|  |  |  |  |  | N/A |
| Very Dissatisfied | Slightly Dissatisfied | Mostly Satisfied | Very Satisfied | Completely Satisfied | ☐ |
| ☐ 1 | ☐ 2 | ☐ 3 | ☐ 4 | ☐ 5 | |

THE ICU

11. How satisfied are you with the atmosphere (mood) of the ICU?
- | | | | | | |
|---|---|---|---|---|--------------------------|
|  |  |  |  |  | N/A |
| Very Dissatisfied | Slightly Dissatisfied | Mostly Satisfied | Very Satisfied | Completely Satisfied | ☐ |
| ☐ 1 | ☐ 2 | ☐ 3 | ☐ 4 | ☐ 5 | |
12. How satisfied are you with your participation in daily rounds?
- | | | | | | |
|---|---|---|---|---|--------------------------|
|  |  |  |  |  | N/A |
| Very Dissatisfied | Slightly Dissatisfied | Mostly Satisfied | Very Satisfied | Completely Satisfied | ☐ |
| ☐ 1 | ☐ 2 | ☐ 3 | ☐ 4 | ☐ 5 | |
13. How satisfied are you with your participation in the care of your critically ill family member?
- | | | | | | |
|---|---|---|---|---|--------------------------|
|  |  |  |  |  | N/A |
| Very Dissatisfied | Slightly Dissatisfied | Mostly Satisfied | Very Satisfied | Completely Satisfied | ☐ |
| ☐ 1 | ☐ 2 | ☐ 3 | ☐ 4 | ☐ 5 | |
14. Some people want everything done for their health problems while others do not want a lot done. How satisfied are you with the LEVEL or amount of health care your family member received in the ICU?
- | | | | | | |
|---|---|---|---|---|--------------------------|
|  |  |  |  |  | N/A |
| Very Dissatisfied | Slightly Dissatisfied | Mostly Satisfied | Very Satisfied | Completely Satisfied | ☐ |
| ☐ 1 | ☐ 2 | ☐ 3 | ☐ 4 | ☐ 5 | |

PART 2: FAMILY SATISFACTION WITH DECISION-MAKING AROUND CARE OF CRITICALLY ILL PATIENTS

INSTRUCTIONS FOR FAMILY OF CRITICALLY ILL PATIENTS

This part of the questionnaire is designed to measure how you feel about YOUR involvement in decisions related to your family member's health care. In the ICU, your family member may have received care from different people. We would like you to think about all the care your family member received when you are answering the questions.

Please check one box that best reflects your views. If the question does not apply to your family member's stay then check the 'Not Applicable' box (N/A)

INFORMATION NEEDS

HOW SATISFIED ARE YOU WITH...

**15. Frequency of Communication
With ICU Doctors?**

How often doctors communicated to you about your family member's condition.



Very
Dissatisfied
☐1



Slightly
Dissatisfied
☐2



Mostly
Satisfied
☐3



Very
Satisfied
☐4



Completely
Satisfied
☐5

N/A
☐

16. Ease of getting information?

Willingness of ICU staff to answer your questions.



Very
Dissatisfied
☐1



Slightly
Dissatisfied
☐2



Mostly
Satisfied
☐3



Very
Satisfied
☐4



Completely
Satisfied
☐5

N/A
☐

17. Understanding of Information?

How well ICU staff provided you with explanations that you understood.



Very
Dissatisfied
☐1



Slightly
Dissatisfied
☐2



Mostly
Satisfied
☐3



Very
Satisfied
☐4



Completely
Satisfied
☐5

N/A
☐

18. Honesty of Information?

The honesty of information provided to you about your family member's condition.



Very
Dissatisfied
☐1



Slightly
Dissatisfied
☐2



Mostly
Satisfied
☐3



Very
Satisfied
☐4



Completely
Satisfied
☐5

N/A
☐

19. Completeness of Information?



	How well ICU staff informed you what was happening to your family member and why things were being done.	Very Dissatisfied ☐ 1	Slightly Dissatisfied ☐ 2	Mostly Satisfied ☐ 3	Very Satisfied ☐ 4	Completely Satisfied ☐ 5	N/A ☐
20.	Consistency of Information? The consistency of information provided to you about your family member's condition (Did you get a similar story from the doctor, nurse, etc.)	 Very Dissatisfied ☐ 1	 Slightly Dissatisfied ☐ 2	 Mostly Satisfied ☐ 3	 Very Satisfied ☐ 4	 Completely Satisfied ☐ 5	N/A ☐
<u>PROCESS OF MAKING DECISIONS</u>							

During your family member's stay in the ICU, many important decisions were made regarding the health care she or he received. From the following questions, select only **one** answer from each of the following set of ideas that best matches your views:

21. **How included or excluded did you feel in the decision making process?**

- ☐ 1 I felt very excluded
- ☐ 2 I felt somewhat excluded
- ☐ 3 I felt neither included nor excluded from the decision making process
- ☐ 4 I felt somewhat included
- ☐ 5 I felt very included

22. **How supported did you feel during the decision making process?**

- ☐ 1 I felt totally unsupported
- ☐ 2 I felt slightly unsupported
- ☐ 3 I felt neither unsupported nor supported
- ☐ 4 I felt supported
- ☐ 5 I felt very supported

23. **Did you feel you had control over the care of your family member?**

- ☐ 1 I felt really out of control and that the health care system took over and dictated the care my family member received
- ☐ 2 I felt somewhat out of control and that the health care system took over and dictated the care my family member received
- ☐ 3 I felt neither in control or out of control
- ☐ 4 I felt I had some control over the care my family member received
- ☐ 5 I felt that I had good control over the care my family member received

24. **When making decisions, did you have adequate time to have your concerns addressed and questions answered?**

- ☐ 1 The time I had was definitely inadequate
- ☐ 2 The time I had was slightly inadequate

- ☐ 3 The time I had was adequate
- ☐ 4 The time I had was more than adequate
- ☐ 5 I had a substantial amount of time

If your family member died during the ICU stay, please answer the following questions (25-27). If your family member did not die, please skip to question 28.

25. Which of the following best describes your views:

- ☐ 1 I felt my family member's life was prolonged unnecessarily
- ☐ 2 I felt my family member's life was slightly prolonged unnecessarily
- ☐ 3 I felt my family member's life was neither prolonged nor shortened unnecessarily
- ☐ 4 I felt my family member's life was slightly shortened unnecessarily
- ☐ 5 I felt my family member's life was shortened unnecessarily

26. During the final hours of your family member's life, which of the following best describes your views:

- ☐ 1 I felt that he/she was very uncomfortable
- ☐ 2 I felt that he/she was slightly uncomfortable
- ☐ 3 I felt that he/she was mostly comfortable
- ☐ 4 I felt that he/she was very comfortable
- ☐ 5 I felt that he/she was totally comfortable

27. During the last few hours before your family member's death, which of the following best describes your views:

- ☐ 1 I felt very abandoned by the health care team
- ☐ 2 I felt abandoned by the health care team
- ☐ 3 I felt neither abandoned nor supported by the health care team
- ☐ 4 I felt supported by the health care team
- ☐ 5 I felt very supported by the health care team

28. Do you have any suggestions on how to make care provided in the ICU better?

29. Do you have any comments on things we did well?

30. Do you have any suggestions on how we could improve the ICU experience for you and your family member?

DEMOGRAPHICS:

Please complete the following to help us know a little about you and your relationship to the patient.

1. I am: ☐ Male ☐ Female ☐ Other

2. I am _____ years old

3. I am the patient's:

- | | | |
|-----------------------------------|----------------------------------|--|
| ☐ Wife | ☐ Husband | ☐ Partner |
| ☐ Mother | ☐ Father | ☐ Sister ☐ Brother |
| ☐ Daughter | ☐ Son | ☐ Other (Please specify): _____ |

4. **Before this most recent event, have you been involved as a family member of a patient in an ICU (Intensive Care Unit)?**

☐ Yes ☐ No

5. **Do you currently live with the patient?** ☐ Yes ☐ No

If no, then on average how often do you see the patient?

☐ More than weekly ☐ Weekly ☐ Monthly ☐ Yearly ☐ Less than once a year

6. **Where do you live?**

☐ In the city or town where the hospital is located ☐ Out of town

7. **Which of the following best describes the highest level of education you have completed?**

- ☐ Did not complete secondary school or high school
- ☐ Completed secondary or high school
- ☐ Had some university education or completed a community college, technical college, or post-secondary program (for example; trade, technical or vocational school)
- ☐ University degree (for example; BA, BSc, BSN)
- ☐ Graduate degree (for example; MD, DDS, DMD, DVM, OD, Master's, or PhD)

We would like to thank you very much for your participation and your opinions.

Supplement IV:

Ethics Paragraph for Qualitative Work

Given the sensitivity of end-of-life care discussions, all qualitative interviews will be conducted by trained personnel with prior experience in grief communication, palliative care interviewing, or social work. Interviewers will undergo training on emotional safety, trauma-informed care, and distress signal recognition. A dedicated emotional risk protocol will be in place. If a participant shows signs of distress during an interview, the interviewer will pause and offer appropriate psychological first aid and/or immediate referral to an in-house counselor or social worker. The research team will also pre-identify high-risk scenarios (e.g., recently bereaved family members, contentious ICU decisions) and offer optional deferral or opt-out at any stage. Informed consent will communicate the voluntary nature of participation and availability of emotional support. All interviews will be conducted in private, empathetic settings, either in person or via secure telecommunication, as per participant preference.

Standard Operating Procedure (SOP)

Title: Qualitative Research Distress Protocol – Participant-Centered Tool for Navigating Distress During Data Collection

Study Title: Implementing a Multifaceted, Comprehensive, Appropriate Care Package for People with Advanced and Terminal Illnesses Needing ICU Support

Principal Investigator: Dr. Naveen Salins

1. Purpose

This SOP outlines procedures to identify, respond to, and manage emotional distress experienced by participants during qualitative research interviews and interactions, particularly when discussing sensitive topics such as ICU experiences and end-of-life care.

2. Scope

This SOP applies to all research staff conducting qualitative data collection, including interviews and focus group discussions with patients' families, healthcare providers, administrators, and public representatives across all project sites.

3. Definitions

- **Distress:** Observable or expressed emotional discomfort or suffering, such as crying, withdrawal, anxiety, or trauma.
- **Participant-Centered Approach:** Ethical and empathetic engagement that prioritizes participant safety, comfort, and autonomy during data collection.

4. Responsibilities

- All qualitative research personnel are responsible for recognizing distress and initiating appropriate actions.
- The Project Research Scientist (Non-Medical) oversees adherence to this protocol and documents all incidents.

5. Procedure

5.1 Pre-Interview Consent and Preparation

- Explain the nature of the interview and potential emotional topics.
- Inform participants of their right to pause, skip questions, or withdraw at any time.
- Provide them with a copy of available mental health resources.

5.2 Recognizing Signs of Distress

Indicators may include:

- Crying or trembling
- Prolonged silence
- Verbal expressions of emotional overwhelm
- Requests to stop the interview

5.3 Responding to Distress

Step 1: Pause and Assess

- Gently interrupt the session.
- Use supportive language: *"Would you like to pause or take a break?"*

Step 2: Supportive Response

- Provide tissues/water.
- Affirm the participant's feelings.
- Ask if they wish to continue or stop the session.

Step 3: Escalation if Required Use the following triage model:

Level	Signs	Action	Responsible
Mild	Tearfulness	Pause and check-in	Interviewer
Moderate	Ongoing crying, anxiety	Stop interview, refer to on-site support	Research Assistant/Nurse
Severe	Panic attack, suicidal thoughts	Immediate referral, document incident	Project Research Scientist

5.4 Documentation and Follow-Up

- Complete a Distress Incident Report Form for all moderate/severe cases.
- Conduct a well-being follow-up call within 48 hours.
- Discuss the incident in team debriefings to improve future responses.

6. Training

All research staff must complete training in:

- Trauma-informed interviewing
- Mental health first aid
- Communication skills in culturally sensitive contexts

7. Confidentiality

Maintain strict confidentiality for all distress events and related documentation. Store reports in a secure, access-controlled folder.

8. Ethical Oversight

The Institutional Ethics Committee will review this protocol, and it will be submitted as part of the study's ethical documentation.

9. Support and Referral Resources

Provide each participant with a contact list including:

- Hospital counselors or social workers
- ICU palliative care team members
- Mental health helpline numbers

10. Appendices

- List of Support Services and Helplines – would be compiled by respective study sites.

Supplement V

Ten key elements of care for the dying patient

Recognition that the patient is dying

Communication with the patient (where possible) and always with family and loved ones

Spiritual care

Anticipatory prescribing for symptoms of pain, respiratory tract secretions, agitation, nausea and vomiting, dyspnoea

Review of clinical interventions should be in the patient's best interests

Hydration review, including the need for commencement or cessation

Nutritional review, including commencement or cessation

Full discussion of the care plan with the patient and relative or carer

Regular reassessment of the patient

Dignified and respectful care after death

Reference: Ellershaw J E, Lakhani M. Best care for the dying patient BMJ 2013; 347 :f4428

doi:10.1136/bmj.f4428

Supplement VI:



The TIDieR (Template for Intervention Description and Replication) Checklist:
Information to include when describing an intervention and the location of the information

Item number	Item	Where located **	
		Primary paper (page or appendix number)	Other † (details)
1.	BRIEF NAME Provide the name or a phrase that describes the intervention.		
2.	WHY Describe any rationale, theory, or goal of the elements essential to the intervention.		
3.	WHAT Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).		
4.	Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.		
5.	WHO PROVIDED For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.		
6.	HOW Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.		
7.	WHERE Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.		

TIDieR checklist

8.	WHEN and HOW MUCH Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.		
9.	TAILORING If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.		
10.*	MODIFICATIONS If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).		
11.	HOW WELL Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.		
12.*	Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.		

** Authors - use N/A if an item is not applicable for the intervention being described. Reviewers - use '?' if information about the element is not reported/not sufficiently reported.

† If the information is not provided in the primary paper, give details of where this information is available. This may include locations such as a published protocol or other published papers (provide citation details) or a website (provide the URL).

* If completing the TIDieR checklist for a protocol, these items are not relevant to the protocol and cannot be described until the study is complete.

* We strongly recommend using this checklist in conjunction with the TIDieR guide (see [BMJ 2014;348:g1687](http://www.bmj.com/2014/348:g1687)) which contains an explanation and elaboration for each item.

* The focus of TIDieR is on reporting details of the intervention elements (and where relevant, comparison elements) of a study. Other elements and methodological features of studies are covered by other reporting statements and checklists and have not been duplicated as part of the TIDieR checklist. When a randomised trial is being reported, the TIDieR checklist should be used in conjunction with the CONSORT statement (see www.consort-statement.org) as an extension of item 5 of the CONSORT 2010 Statement. When a clinical trial protocol is being reported, the TIDieR checklist should be used in conjunction with the SPIRIT statement as an extension of item 11 of the SPIRIT 2013 Statement (see www.spirit-statement.org). For alternate study designs, TIDieR can be used in conjunction with the appropriate checklist for that study design (see www.tidier.org).

TIDieR checklist

Adapted from –

Hoffmann T, Glasziou P, Boutron I, Milne R, Perera R, Moher D, Altman D, Barbour V, Macdonald H, Johnston M, Lamb S, Dixon-Woods M, McCulloch P, Wyatt J, Chan A, Michie S. Better reporting of interventions: template for intervention description and replication (TIDieR) checklist and guide. *BMJ*. 2014;348:g1687.

Supplement VII

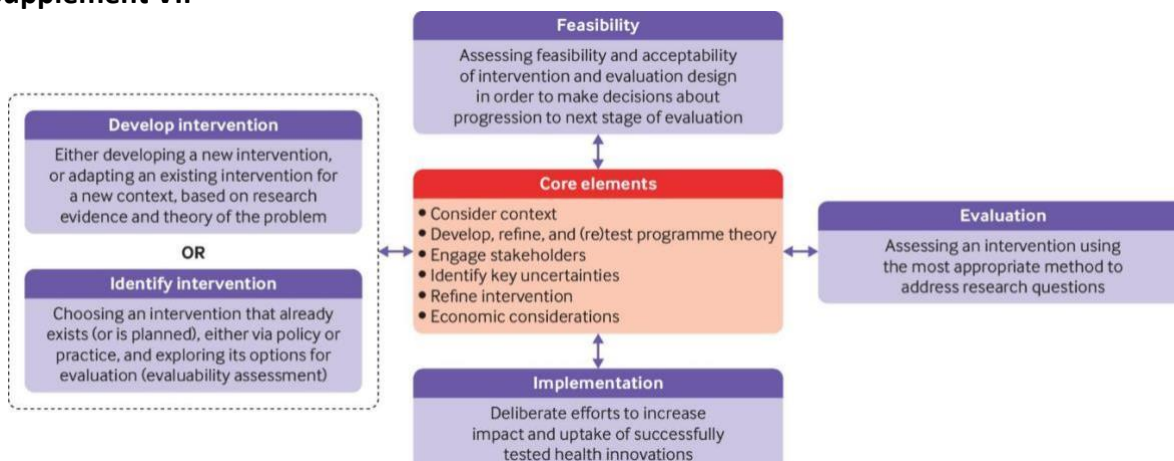


Fig 1 Framework for developing and evaluating complex interventions. Context=any feature of the circumstances in which an intervention is conceived, developed, evaluated, and implemented; programme theory=describes how an intervention is expected to lead to its effects and under what conditions—the programme theory should be tested and refined at all stages and used to guide the identification of uncertainties and research questions; stakeholders=those who are targeted by the intervention or policy, involved in its development or delivery, or more broadly those whose personal or professional interests are affected (that is, who have a stake in the topic)—this includes patients and members of the public as well as those linked in a professional capacity; uncertainties=identifying the key uncertainties that exist, given what is already known and what the programme theory, research team, and stakeholders identify as being most important to discover—these judgments inform the framing of research questions, which in turn govern the choice of research perspective; refinement=the process of fine tuning or making changes to the intervention once a preliminary version (prototype) has been developed; economic considerations=determining the comparative resource and outcome consequences of the interventions for those people and organisations affected

Adapted from –

Skivington K, Matthews L, Simpson SA, Craig P, Baird J, Blazeby JM, Boyd KA, Craig N, French DP, McIntosh E, Petticrew M, Rycroft-Malone J, White M, Moore L. A new framework for developing and evaluating complex interventions: update of Medical Research Council guidance. BMJ. 2021 Sep 30;374:n2061. doi: 10.1136/bmj.n2061. PMID: 34593508; PMCID: PMC8482308.

Supplement VIII**Sample size calculation details for objective 2:**

- To determine the appropriate sample size for the pilot study (Objective 2), we performed a power analysis based on the need to detect a medium effect size in clinical outcomes such as symptom relief, family satisfaction, or the quality of dying. We used Cohen's $w = 0.3$, representing a moderate effect size suitable for categorical outcomes assessed via chi-square tests for this pilot. Assuming a power of 80% ($1-\beta = 0.80$) and a significance level of $\alpha = 0.05$, the required minimum sample size per group was calculated to be 87 participants for chi-square tests

$$n = \frac{(Z_{1-\alpha/2} + Z_{1-\beta})^2}{w^2}$$

Where:

- $Z_{1-\alpha/2} = 1.96$ (for $\alpha = 0.05$)
- $Z_{1-\beta} = 0.84$ (for power = 0.8)
- $w = 0.3$

$$n = \frac{(1.96 + 0.84)^2}{0.3^2} = \frac{7.84}{0.09} \approx 87$$

- This sample size is appropriate for detecting clinically meaningful differences while maintaining feasibility in a pilot setting. The total number of ICU patients required for the pilot phase, comparing two models of ICU care (e.g., Open vs Closed ICUs), is thus 174 patients. Considering approximately 10 % loss in patient numbers, it comes to 100 per clinical site. These will be further distributed across four clinical subgroups: terminal cancer, end-stage organ failure, progressive neurological diseases, and post-trauma/neurosurgical conditions, to allow subgroup-level feasibility and outcome analysis.
- This allows subgroup analysis with ~25 patients per group in each phase—statistically justifiable.

Sample size calculation for Objective 3:

- Primary Clinical Outcome: *Improvement in QODD score (Quality of Dying in ICU)*.
- Based on prior studies:
 - Mean QODD score ≈ 60 (SD ~ 20)
 - MCACP is expected to improve it by 8–10 points

Sample Size Formula for Comparing Two Means (Pre vs Post)

Using the formula for comparing means in a pre-post quasi-experimental study:

Using the formula for comparing means in a pre-post quasi-experimental study:

$$n = \frac{2 \cdot (Z_{\alpha/2} + Z_{\beta})^2 \cdot \sigma^2}{\delta^2}$$

Where:

- $Z_{\alpha/2} = 1.96$ for 95% confidence
- $Z_{\beta} = 0.84$ for 80% power
- $\sigma = 20$ (standard deviation of QODD score)
- $\delta = 8$ (expected difference in mean score)

So, to detect an 8-point improvement in QODD score (with SD = 20) at 95% confidence and 80% power, we need:

- Minimum: 98 patients per group (pre & post)
- That's 196 patients per site total. Considering a 30 % loss in subject data, we will attempt to recruit 150 patients pre + 150 patients post per site
- Across 4 patient groups

That gives:

- 300 patients $\times$ 4 ICUs = 1200 total per phase
- Ensures adequate power not just overall but also for subgroup analysis across the 4 patient types:
 - Advanced Cancer

- Organ Failure
- Neurological Disease
- ICU-related Terminal Conditions
- This allows subgroup analysis with ~75 patients per group in each phase—statistically justifiable.

References

Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Lawrence Erlbaum Associates.

Julious, S. A. (2005). Sample size of 12 per group rule of thumb for a pilot study. Pharmaceutical Statistics, 4(4), 287–291.

Browne, R. H. (1995). On the use of a pilot sample for sample size determination. Statistics in Medicine, 14(17), 1933–1940. <https://doi.org/10.1002/sim.4780141709>

Supplement IX:
Inclusion Criteria

General Criteria (at least one of the below) • Age $\geq$ 65 years • Frailty – Clinical Frailty Score $\geq$ 7 • Poor performance status – AKPS 40 or below • Two or more co-morbid illnesses • One or more end-stage organ impairment (pulmonary, cardiac, renal, hepatic etc.) • Stage 4 Cancer or Relapsed/Recurrent Cancers
In patients admitted to an Intensive Care Unit having (at least one of the below) • Progressively worsening multi-organ dysfunction not responding to a reasonable duration of aggressive medical interventions • Coma due to causes with non-reversible effects like major infarcts, intracranial bleed or traumatic brain injury • Metastatic malignancies where there are no treatment options or where the treatment options cannot meet the goals of care • Post-cardiac arrest status with poor neurological outcomes • Chronic neurological conditions where there is a progressive deterioration in the cognition and functional ability leading to a poor quality of life like progressive dementia, chronic vegetative state, quadriplegia • Any condition where the physician predicts a low probability of meaningful survival • Extreme drug-resistant infections
In patients with Neurological Conditions • Advanced cases of Alzheimer's disease with total dependence (Dementia Rating Scale 50 or above) • Stroke survivors with GCS <4 after three days of absent corneal reflex • Multiple Sclerosis EDSS scale 9.5 (Confined to bed and totally dependent. Unable to communicate effectively or eat/swallow) • ALS rating scale (<6) • Parkinson's Disease UPDRS Scale <150 • CNS Infections like meningitis and encephalitis with progressive worsening, multiorgan dysfunction that does not respond to a period of aggressive therapy • Post-cardiac arrest status with poor neurological outcomes • Any condition where the physician predicts a low probability of meaningful survival • Supra refractory status epilepticus after three days of aggressive management.
In patients with Neuro-Surgical Conditions • Severe traumatic brain injury (GCS < 6) with dilated fixed pupils > 48 hours • High cervical spinal cord injury with quadriplegia and absence of any spontaneous respiratory effort > 7 days • Patients with ICH score > 5 for more than 48 hours

- Patients with Malignant ischemic infarct (pre and post-operative) with GCS < 6 and fixed dilated pupils
- Recurrent high-grade brain tumours with GCS < 6 and poor KPS < 70
- Metastatic brain disease with GCS < 6 and poor KPS < 70
- Patients with diffuse hypoxic injury secondary to any aetiology
- Congenital paediatric developmental abnormalities incompatible with prolonged survival

In patients admitted to the medical wards

- Terminally ill patient with progressive multi organ failure needing life support systems and medicines (sepsis and MODS) not showing signs of response to therapy
- Advanced progressive organ failure needing life support system and medicine (renal, liver, cardiac, neurological or respiratory failure) and not showing signs of response to therapy and organ transplantation or replacement therapy is not feasible
- Advanced malignancy on palliation with additional complications like severe infection, bleeding or organ failure

In patients with cancer

- Metastatic cancer or advanced cancer with no scope of further disease modifying treatment
- Expected survival weeks to days
- Poor performance status and needing significant assistance with activities of daily living
- Multiple unplanned admissions and ER visits in the last 12 months